Diffusion Models on Manifolds

Abstract

Sections 4, 5 and 6 are from [1]. Specifically, these sections are based on Sections 5.3 and 5.4. Section 7 is based on the lecture Markov Processes by Prof. Slowik. Sections 8 and 9 are based on [2]. Here Section 7.4 and Chapter 9 were used. Section 10 is based on Chapter 5 and section 7.1 of [3]. The goal is to first derive all technical backround needed to understand [4]. This paper is discussed in the final section.

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SECTION 1

MOTIVATION AND WHY DO WE WANT TO USE DIFFUSION

A standard diffusion

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t$$

is like a drop of ink in water. As particles move forward in time they diffuse. If one wants to go backward in time, one needs a process that moves back to that initial single drop. There must be a force/drift in the backward process that pulls the trajectories back to their initial condition. We will see under assumptions on the forward diffusion process that the backward dynamics $\bar{X}_t := X_{T-t}$

$$d\bar{X}_t = (-b(\bar{X}_t + \sigma(\bar{X}_t)^2 \nabla \log(p(T-t, \bar{X}_t))dt + \sigma(\bar{X}_t)d\tilde{W}_t, \quad X_t \sim p(t, \cdot),$$

where $\tilde{W}_t$ is the Brownian motion wrt. the reverse filtration, is a Markov process, although the drift plays the key role and is different to the forward process. It depends on $\nabla \log(p(t,x))$, although $p(t,x)$ is the density of the law of the forward process. We call that the score function. If one chooses the initial condition of the forward dynamics $X_0 \sim \mathbb{P}$ of the distribution one wants to estimate/learn, then under certain assumption of the drift b and the diffusion σ, the process will converge to the standard normal distribution, i.e.

$$X_\infty \sim \mathcal{N}(0, Id).$$

Intuitively, we take a sample from the distribution we want to learn, diffuse it, so that only gaussian noise remains. We will later see that, that the reverse diffusion process with standard normal initial distribution converges to the initial condition of the forward diffusion process, i.e. the distribution we want to learn. The goal is to learn $\nabla \log(p(t,x))$, as this is the relevant part of the reverse process. The idea is to estimate the score function by parameterizing the score function $s_\theta(t,x)$ by approximating it using Neural Networks. One may consider

$$\mathbb{E}^{t\sim\mathcal{U}([0,T])}[\mathbb{E}^{X_t\sim p(t,\cdot)}[\|\nabla \log(p(t,x) - s_\theta(t,x)\|_2^2]] = \min_{\theta\in\Theta}!$$

However, $p(t,x)$ is unknown, because

$$p(t,x) \overset{???}{=} \int_{\mathbb{R}^d} p(t,x \mid z)p_0(z)dz,$$

which requires the unknown $p_0 = d\mathbb{P}^{X_0}$. One can show (???) that the above is objective is equivalent to

$$\mathbb{E}^{t\sim\mathcal{U}([0,T])}[\mathbb{E}^{X_0\sim\mathbb{P}^{X_0}}[\mathbb{E}^{X_t\sim p(t,\cdot|X_0)}[\|\nabla \log(p(t,x \mid X_0) - s_\theta(t,x)\|_2^2]]] = \min_{\theta\in\Theta}!$$

SECTION 2

STOCHASTIC PROCESSES RECAP

Theorem 2.1

Let $(X^n)_{n\in\mathbb{N}}$ be a tight sequence of continuous processes such that

$$\forall 0 \le t_1 < ... < t_d < \infty : \quad (X^n_{t_1}, ..., X^n_{t_d})_{n\in\mathbb{N}}$$

converges in distribution, i.e.

$$w - \lim_{n\to\infty} \mathbb{P} \circ (X^n_{t_1}, ..., X^n_{t_d})^{-1} = \lim_{n\to\infty} \int_{\mathbb{R}^d} df \mathbb{P} \circ (X^n_{t_1}, ..., X^n_{t_d})^{-1}, \quad \forall f : \mathbb{R}^d \to \mathbb{R} \text{ bounded and continuous}$$

exists. Then

- $w - \lim_{n\to\infty} \mathbb{P}_n := w - \lim_{n\to\infty} \mathbb{P} \circ (X^n_{t_1}, ..., X^n_{t_d})^{-1} = P$
- On P the coordinate mapping process $W_t(\omega) = \omega(t)$ on $C([0,\infty))$ satisfies

$$w - \lim_{n\to\infty} \mathbb{P} \circ (X^n_{t_1}, ..., X^n_{t_d})^{-1} = P \circ (W_{t_1}, ..., W_{t_d})^{-1}, \quad \forall 0 \le t_1 < ... < t_d < \infty.$$

Proof. Let $(X^{n_m})_{m\in\mathbb{N}}$ be a subseqeuence of $(X^n)_{n\in\mathbb{N}}$. Because this subseqeunce is again tight (subsequences can be chosen arbitrarly) it has another subsequence $(X^{n_{m_l}})_{l\in\mathbb{N}}$ so that

$$w - \lim_{l\to\infty} \mathbb{P} \circ (X^{n_{m_l}})^{-1} = P.$$

Assume there is another subsequence $(X^{n_{m_k}})_{k\in\mathbb{N}}$ that induces a measure $C([0,\infty))$ with

$$Q := w - \lim_{k\to\infty} \mathbb{P} \circ (X^{n_{m_k}}).$$

Then for any $\pi : C([0,\infty)) \to \mathbb{R}^d$ we have

$$P \circ \pi^{-1} = Q \circ \pi^{-1},$$

yielding $Q = P$, which implies the weak convergence of $(P^n)_{n\in\mathbb{N}}$ to P.
Suppose $(\mathbb{P}^n)_{n\in\mathbb{N}}$ did not converge weakly to P, then $\exists f : C([0,\infty)) \to \mathbb{R}$, bounded and continuous such that $\int f d\mathbb{P}_n$ does not exist or does not equal $\int f dP$. In both cases choose a subsequence $(\mathbb{P}^{n_k})_{k\in\mathbb{N}}$ with $\lim_{k\to\infty} \int df d\mathbb{P}^{n_k}$ does not exists or does not equal $\int f dP$. Any further subseqeunce also cannot converge weakly to P. However, in the abstract before it was shown that

$$\lim_{l\to\infty} \int f d\mathbb{P}^{n_{k_l}} = \int f dQ = \int f \, dP,$$

which implies that $(\mathbb{P}^{n_k})_{k\in\mathbb{N}}$ should converge weakly to P, which contradicts the construction before. □

2.1 Canonical Construction of Stochastic Processes

Construction of a stochastic process on I with $|I| = \infty$.

- Define a σ-algebra $\mathcal{E}^{\otimes I}$ on $E^I := \{f : I \to E\}$
- Understand all probability measures on $\mathcal{E}^{\otimes I}$
- Reinterpret stochastic processes as path valued random variables.

For $d < \infty$ the product σ-algebra

$$\mathcal{E}^{\otimes d} := \mathcal{E} \otimes \cdots \otimes \mathcal{E} := \sigma(\{B_1 \times \cdots \times B_d \mid B_i \in \mathcal{E}\}),$$

where $B_1 \times \cdots \times B_d$ are the rectangular sects.

Remark 2.1

If $\mathcal{E} = \sigma(\mathcal{A})$ then

$$\mathcal{E}^{\otimes d} = \sigma(\{B_1 \times \cdots \times B_d \mid B_i \in \mathcal{A}\}).$$

???

We now want to generalize to infinite products.

Definition 2.2

A subset $C \subset E^I$ is called finitely generated cylinder set if

$$C = \pi_J^{-1}(B_1 \times \cdots \times B_n) = \{f : I \to E \mid f(t_1) \in B_1, ..., f(t_n) \in B_n\},$$

where $J := \{t_1, ..., t_n\} \subseteq I$, $B_1, ..., B_n \in \mathcal{E}$ and the finite projection

$$\pi_J : E^I \to E^{|J|}, f \mapsto \pi_J(f) := (f(t_1), ..., f(t_n)).$$

We say the set C of paths is spanned by finitely many time points $t_1, ..., t_n$ and the finite dimensional cylinder $B_1 \times \cdots \times B_n \in \mathcal{E}^{\otimes d}$.

Definition 2.3

The path σ-algebra $\mathcal{E}^{\otimes I}$ on E^I is the smallest σ-algebra that contains all finitely generated cylinder sets. In the case $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ one writes $\mathcal{B}(\mathbb{R}^d), \mathcal{B}(\mathbb{R}^\infty), \mathcal{B}(\mathbb{R}^{[0,T]})$.

A σ-algebra defined using finitely generated cylinder sets can only at most cover countably many restrictions. We call a set finitely generated if there exists $J \subseteq I$ (finite or countable) such that

$$f \in C \quad \iff \quad f\big|_C \in B, \quad B \in \mathcal{E}^{\otimes J}.$$

Note that here we do not only use cylinder sets $B_1 \times \cdots \times B_d$, as in the next proposition shows that we need more than these cylinders sets to form a σ-algebra.

Proposition 2.4

The set

$$\mathcal{V} := \{C \subseteq E^I \mid \exists J \subseteq I \text{ countable}, B \in \mathcal{E}^{\otimes J}, \forall f \in E^I : \quad f \in C \iff f\big|_J \in B\}$$

of all countably generated sets is a σ-algebra on E^I that contains $\mathcal{E}^{\otimes I}$.

Example 2.1

If $E = \mathbb{R}$ and $I \subseteq \mathbb{R}$ such that $|I| = \infty$, then $\{f\}, \{f \equiv c \in E\}, \{f : I \to E \mid f \geq 0\}, C(I, E)$ are not measurable sets. If any of those sets were measurable, then that set should be with the above proposition in $\mathcal{V}$ as well. This means that there should exist $J \subseteq I$ countable and $B \in \mathcal{E}^{\otimes J}$ so that any $f\big|_J \in B$, meaning for $f \in C$, we only need finite many points to be inside a measurable set of E^I. This however fails the definition of these sets: Choose $i \notin J$ and set $f(i)$ in such a way, that it does not satisfy the defining property anymore, e.g. a discontinuous point for example. Then f is no longer in C even though it is measurable. So statements of the $\mathbb{P}$-a.s. kind need to be done with caution.

We now construct the measures on $\mathcal{E}^{\otimes I}$. The idea is as follows

- Downsize $\mathbb{P}$ to only finite cylinder sets (that are $\cap$-stable). This leads to finite-dimensional distributions
- Check that $\{\mathbb{P}_J \mid |J| < \infty\}$ satisfies a natural property, which will be consistency.
- Show that for these finite dimensional consistent families there exists an infinite dimensional probability measure.

Definition 2.5

If $\mathbb{P}$ is a probability measure on $\mathcal{E}^{\otimes I}$ and $J \subseteq I$ is finite, then the unique probability measure on $\mathcal{E}^{\otimes J}$ is given by

$$\mathbb{P}_J(B_1 \times \cdots \times B_{|J|}) := \mathbb{P}(\pi_J^{-1}(B_1 \times \cdots \times B_{|J|})), \quad B_i \in \mathcal{E},$$

is called finite-dimensional marginal distribution of $\mathbb{P}$.

On the one hand we have the boxes, i.e. $B_1, ..., B_n \in \mathcal{E}$ and cylinder sets $\{f : I \to E \mid f(t_1) \in B_1, ..., f(t_n) \in B_n\}$. The finite-dimensional marginals are the measures that we get by crossing out the functions away from J, meaning it is the distribution of $\pi_J(f)$ on these boxes.

Proposition 2.6

Every probability measure $\mathbb{P}$ on $\mathcal{E}^{\otimes I}$ is uniquely determined through the family of all finite-dimensional marginal distributions $\{\mathbb{P}_J \mid J \subseteq I \text{ finite }\}$.

The next result will lead to the characterizing property of finite dimensional families related to measures on $\mathcal{E}^{\otimes I}$.

Proposition 2.7

If $\mathbb{P}$ is a probability measure on $\mathcal{E}^{\otimes I}$ then the family of all finite-dimensional marginal distributions $\{\mathbb{P}_J \mid J \subseteq I \text{ finite }\}$ satisfies for $\pi_{J,J'} : E^J \to E^{J'}$

$$\forall J' \subseteq J \subseteq I \text{ finite with } \pi_{J,J'}(f) = f\big|_{|J|} : \quad \mathbb{P}_J \circ \pi_{J,J'}^{-1} = \mathbb{P}_{J'}.$$

Definition 2.8

A set of finite-dimensional distributions $\{\mathbb{P}_J \mid |J| < \infty\}$ on I is called consistent if

$$\forall J' \subseteq J : \quad \mathbb{P}_J \circ \pi_{J,J'}^{-1} = \mathbb{P}_{J'}.$$

This means "adding" E's to a product $B_1 \times \cdots \times B_n \times E$ does not change the measure. For the infinite dimensional distribution we do not need more properties for the existence of the infinite dimensional distribution.

Theorem 2.9

Let E be a polish metric space, $\mathcal{E} = \mathcal{B}(E)$ and I an index set. If $\{\mathbb{P}_J \mid J \subseteq I \text{ finite}\}$ is a consistent family then there exists a unqiue $\mathbb{P}$ on $\mathcal{E}^{\otimes}$ such that

$$\forall J \subseteq I \text{ finite:} \quad \mathbb{P} \circ \pi_J^{-1} = \mathbb{P}_J.$$

How should we know $\mathbb{P}_J$ without knowing $\mathbb{P}$?

Definition 2.10

Suppose $(E, \mathcal{E})$ is a measurable space and $\mathbb{P}$ a probability measure on $\mathcal{E}$. Then, a measure μ on $\mathcal{E}^{\otimes \mathbb{N}}$ is called infinite dimensional product measure if

$$\forall n \in \mathbb{N}, B_i \in \mathcal{E}: \quad \mu(B_1 \times \cdots \times B_n \times E \times \cdots) = \mathbb{P}(B_1) \cdots \mathbb{P}(B_n).$$

We typically write for an infinite product measure $\mathbb{P}^{\otimes \infty}$ or $\mathbb{P}^{\otimes \mathbb{N}}$.

Theorem 2.11

If E is polish metric space and $\mathbb{P}$ a probability measure on $\mathcal{B}(E)$, then the infinite dimensional product measure $\mathbb{P}^{\otimes \infty}$ exists and is unique.

Next, we want to understand the connection of stochastic processes and path valued random variables.

Proposition 2.12

$X := (X_t)_{t \in I}$ is E-valued stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$ if and only if

$$X : \omega \mapsto (t \mapsto X_t(\omega))$$

is a path valued random variable.

Proof. "$\Leftarrow$"
For $B \in \mathcal{E}$

$$X_t^{-1}(B) = X^{-1}(\pi_t^{-1}(B)) \in \mathcal{A},$$

because $\pi_t^{-1}(B)$ is a finitely generated cylinder set and X is $(\mathcal{A}, \mathcal{E}^{\otimes I})$-measurable.
"$\Rightarrow$"
We only need to check measurability for an arbitrary generator. We use the cylinder sets $\pi_t^{-1}(B)$. Then

$$X^{-1}(\pi_t^{-1}(B)) = \{\omega \in \Omega \mid X_t(\omega) \in B\} = X_t^{-1}(B) \in \mathcal{A}.$$

□

Definition 2.13

Suppose X is an E-valued stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$, then

- The image measure

$$\mathbb{P}^X(A) := \mathbb{P}(X \in A), \quad A \in \mathcal{E}^{\otimes I},$$

is called law of X.

- For $J := \{t_1, ..., t_n\} \subseteq I$ we call

$$\mathbb{P}^X_{t_1,...,t_n} \sim (X_{t_1}, ..., X_{t_n})$$

is finite dimensional distribution (fdd) of X.

Usually, the law does not play a big role and we exclusively deal with the fdds.

Proposition 2.14

The law of a stocahstic process X is uniquely determined by all its finite-dimensional distributions

$$\mathbb{P}^X_{t_1,...,t_n} = \mathbb{P}(X_{t_1} \in \cdot, ..., X_{t_n} \in \cdot).$$

Definition 2.15

Two stochastic processes X and Y with index set I have the same law if $\mathbb{P}^X = \mathbb{P}^Y$. We say they have the same finite-dimensional distribution if

$$\forall \{t_1, ..., t_n\} \subseteq I : \quad \mathbb{P}^X_{t_1,...,t_n} = \mathbb{P}^Y_{t_1,...,t_n}$$

and write $\mathbb{P}^X \overset{fdd}{=} \mathbb{P}^Y$.

Theorem 2.16

Suppose E is a polish metric space, I an index set and $\{\mathbb{P}_J \mid J \subseteq I \text{ finite}\}$ a consistent family of finite-dimensional distributions, then there exists a stochastic process X on $(\Omega, \mathcal{F}, \mathbb{P})$ so that

$$\forall J \subseteq I : \quad \mathbb{P}^X_J = \mathbb{P}_J.$$

One tries to avoid the canonical construction, because statements such as "X is $\mathbb{P}$-a.s. continuous/..." are problematic on the canonical space. This is why we now want to derive continuous modifications. For that we need the stochastic process to be a random variable in a polish space. First, we consider the path space of continuous functions and stochastic processes with continuous oaths. Second, we show how to construct these. We cannot use Kolmogorov's extension theorem, as it breaks down for continuous functions. Note that it is already known that $(C([0,T]), \|\cdot\|_\infty)$ is a polish space.

Proposition 2.17

- $$d(f,g) := \sum_{i=1}^{k} \frac{d : k(f,g) \vee 1}{2^k}, \quad d_k(f,g) := \sup_{x \in [0,k]} |f(x) - g(x)|,$$

is a metric on $C([0,\infty))$.

- $\lim_{n\to\infty} d(f_n, g) = 0 \iff \lim_{n\to\infty} \sup_{x\in K} |f_n(x) - g(x)| = 0$ for every K compact.
- $(C([0,\infty)), d)$ is a polish metric space.

The product σ-algebra for $\mathbb{R}^{[0,\infty)}$ has no structure, however, the Borel σ-algebra generated by open sets

for $C([0,\infty))$ has more, because for the product σ-algebra can only have measurable sets, that have finite restrictions. Because the continuous functions are seperable, there are not any for the Borel σ-algebra on open sets.
Cylinder sets of continuous functions are also the generators now:

Proposition 2.18

$\mathcal{B}(C([0,\infty)))$ is generated by cylinder sets of continuous functions, i.e.

$$\begin{aligned}\mathcal{B}(C([0,\infty))) &= \sigma(\{\pi_t^{-1}(B) \mid t \geq 0, B \in \mathcal{B}(\mathbb{R})\}) \\ &= \sigma(\{\text{ finitely generated cylinder sets of continuous functions}\}),\end{aligned}$$

where $\pi_t : C([0,\infty)) \to \mathbb{R}, f \mapsto f_t$.

Because $\mathcal{B}(C([0,\infty)))$ has the same type of generator as $\mathcal{E}^{\otimes I}$ we can copy ideas for probability measures from before. If $\mathbb{P}$ is a probability measure on $\mathcal{B}(C([0,\infty)))$, then for every $t_i \geq 0$ and $B_i \in \mathcal{B}(\mathbb{R})$ we have

$$\mathbb{P}_{t_1,\dots,t_n}(B_1 \times \cdots \times B_k) := \mathbb{P}(\{f \in C([0,\infty)) \mid f(t_1) \in B_1, \dots, f(t_k) \in B_k\}),$$

is called the finite dimensional distribution. Since this probability measure is defined on a $\cap$-stable generator, then finite dimensional distribution uniquely determines the law.

Corollary 2.19

A probability measure $\mathbb{P}$ on $\mathcal{B}(C([0,\infty)))$ is uniquely determined by the family $\{\mathbb{P}_{t_1,\dots,t_k} \mid k \in \mathbb{N}, t_i \geq 0\}$ of all finite-dimensional distributions.

Proposition 2.20

$\mathcal{B}(\mathbb{R}^{[0,\infty)})\big|_{C([0,\infty))} = \mathcal{B}(C([0,\infty)))$.

we cannot use Kolmogorovs extension theorem to construct measures on $\mathcal{B}(C([0,\infty)))$.

Definition 2.21

Suppose X is a stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$ then

- X is continuous/differentiable/... if $\omega \mapsto X_t(\omega)$ is continuous/differentiable/... for all $\omega \in \Omega$.
- X is *a.s.* continuous/differentiable/... if there exists $C \in \mathcal{F}$ with $\mathbb{P}(C) = 1$ such that $t \mapsto X_t(\omega)$ is continuous/differentiable/... for every $c \in C$.

We have already defined the law $\mathbb{P}^X(B) := \mathbb{P}(X \in B)$ on the product σ-algebra. It would be nice to define it on $C([0,\infty))$ as well. This is why we need $X : \Omega \to C([0,\infty)) \subset \mathbb{R}^{[0,\infty)}$ to be $(\mathcal{F}, \mathcal{E}^{\otimes I}$ and $(\mathcal{F}, \mathcal{B}(C([0,\infty))))$ measurable. This is because if $B \in \mathcal{B}(C([0,\infty)))$ then there exists $B' \in \mathcal{E}^{\otimes I}$ with $B = B' \cap C([0,\infty))$ and it holds

$$X^{-1}(B) = X^{-1}(B) \cap X^{-1}(C([0,\infty))) = X^{-1}(B') \cap \Omega \in \mathcal{F}.$$

Hence, introducing continuous processes can be interpreted as random variables taking values in $C([0,\infty))$ and inducing a law on $\mathcal{B}(C([0,\infty)))$.

Definition 2.22

If X is a continuous stochastic process then the law $\mathbb{P}^{X,c}$ of X on $\mathcal{B}(C([0,\infty)))$ is

$$\mathbb{P}^{X,c}(B) := \mathbb{P}(X \in B), \quad B \in \mathcal{B}(C([0,\infty))).$$

We now construct the continuous stochastic processes.

Definition 2.23

A function $f : [0,\infty) \to \mathbb{R}$ is called Hölder continuous with index $\gamma \in (0,1]$ and constant K if

$$\forall s,t \in \mathbb{R}: \quad |f(t) - f(s)| \leq k|t-s|^{\gamma}.$$

We call f Hölder continuous on compacts on $\gamma \in (0,1]$ with $(K_n)_{n\in\mathbb{N}}$ if

$$\forall s,t \in \mathbb{N}: \quad |f(t) - f(s)| \leq K_N|t-s|^{\gamma},$$

for all $N \in \mathbb{N}$. f is locally Hölder continuous of index $\gamma \in (0,1]$ if

$$\forall t \in \mathbb{N} \exists \mathcal{U}_t \subseteq \mathbb{N}, \text{ with } t \in \mathcal{U}_t \forall s \in \mathcal{U}_t: \quad |f(t) - f(s)| \leq K_t|t-s|^{\gamma}.$$

f is Hölder continuous in t of index $\gamma \in (0,1]$ if

$$\exists \mathcal{U}_t \ni t \in \mathbb{N} \text{ and } K_t: \quad |f(t) - f(s)| \leq K_t|t-s|^{\gamma}, \quad s \in ???.$$

Definition 2.24

Let X and $\tilde{X}$ be a stochastic process with index set I on $(\Omega,\mathcal{F},\mathbb{P})$. Then $\tilde{X}$ is a modification of X if

$$\forall t \in I: \quad \mathbb{P}(X_t = \tilde{X}_t) = 1.$$

Definition 2.25

Let X and $\tilde{X}$ be a stochastic process with index set I on $(\Omega,\mathcal{F},\mathbb{P})$. Then X and $\tilde{X}$ are called indistinguishable if there exists $C \in \mathcal{F}$ with $\mathbb{P}(C) = 1$ so that for every $\omega \in \Omega$: $X_t(\omega) = \tilde{X}_t(\omega)$.

Modifications of erach other is stronger than equality in law. Indistinguishable is the strongest here.

Example 2.2

Suppose $U \sim \mathcal{U}([0,1])$ on $(\Omega,\mathcal{F},\mathbb{P})$ and define $X_t := 0$ and $\tilde{X}_t := \mathbf{1}_U(f)$ on $I = [0,1]$. Then $\tilde{X}$ is a modification of X, because

$$\mathbb{P}(X_t = \tilde{X}_t) = \mathbb{P}(U \neq t) = 1.$$

However, their paths are a.s. different.

Theorem 2.26

Suppose X is a stochastic process on $(\Omega,\mathcal{F},\mathbb{P})$ such that for all $T > 0$ exists c, α, β

$$\forall s,t \leq T: \quad \mathbb{E}[|X_t - X_s|^{\alpha}] \leq C|t-s|^{1+\beta}.$$

Then there exists $\tilde{X}$ a continuous modification of X s.t. $\tilde{X}$ has Hölder continuous paths of order $\gamma < \frac{\beta}{\alpha}$.

Finally, we now consider weak convergence of continuous processes

Proposition 2.27

Let $\mu, \mu^1, ...$ be sequence of probability measures on $\mathcal{B}(C([0,\infty)))$, then $\mu^n \overset{weak}{\to} \mu$, $n \to \infty$ if and only if

- $\{\mu^n \mid n \in \mathbb{N}\}$ is tight in $\mathcal{M}_1(C([0,\infty)))$
- All finite-dimensional distributions converge, i.e.

$$\forall J \subseteq [0,\infty) \text{ finite:} \quad \mu_J^n := \mu^n \circ \pi_J^{-1} \overset{weak}{\to} \mu \circ \pi_J^{-1} = \mu_J, \quad n \to \infty.$$

We are mostly interested in the stochastic reformulation and not in the abstract measure theory.

Definition 2.28

Suppose $X, X^1, ...$ are stochastic processes with *a.s.* continuous paths and laws $\mathbb{P}^X, \mathbb{P}^{X^1}, ...$ on $C([0,\infty))$.

- $(X^n)_{n\in\mathbb{N}}$ converges weakly to X if $\mathbb{P}_{X^n} \overset{weak}{\to} \mathbb{P}_X$, $n \to \infty$. We write $X^n \Rightarrow X$.
- The finite-dimensional distributions (fdds) of $(X^n)_{n\in\mathbb{N}}$ converge to those of X if for every $0 \le t_1 \le ... \le t_n$

  $$(X_{t_1}^n, ..., t_{t_k}^n) \overset{(d)}{\to} (X_{t_1}, ..., X_{t_k}), n \to \infty$$

  or equivalently

  $$\mathbb{P}^{X^n}_{t_1,...,t_k} \overset{weak}{\to} \mathbb{P}_{t_1,...,t_k}, n \to \infty.$$

  We write $X \overset{fdd}{\Rightarrow} X$.

Remark 2.2

$$X^n \Rightarrow X, \quad n \to \infty \quad \overset{cont.MappingThm}{\Rightarrow} \quad X^n \overset{fdd}{\Rightarrow} X, n \to \infty.$$

The continuous mapping theorem yields that weak convergence implies that

$$\forall F : C([0,\infty)) \to \mathbb{R} : \quad F(X^n) \overset{(d)}{=} F(X),$$

whereas convergence in ffd only implies that

$$\forall F : \mathbb{R}^k \to \mathbb{R} \text{ continuous:} \quad F(X_{t_1}^n, ..., t_{t_k}^n) \overset{(d)}{=} F(X_{t_1}, ..., X_{t_k}).$$

Theorem 2.29

A sequence $(X^n)_{n\in\mathbb{N}}$ of continuous stochastic processes weakly converges to X if and only if

- $\{\mathbb{P}^{X^n} \mid n \in \mathbb{N}\} \subseteq \mathcal{M}_1(C([0,\infty)))$ is tight

- $X^n \overset{ffd}{\Longrightarrow} X$, $n \to \infty$.

Recall A is called relatively compact if all sequences have convergent subsequences in A. It is relatively compact if all sequences have convergent subsequences with the limit not needing to belong to A.

Remark 2.3

Ascela-Ascoli: $A \subseteq C([0,\infty))$ is relatively compact if and only if

- $\{f(0) \mid f \in A\} \subseteq \mathbb{R}$ is bounded
- For every $N \in \mathbb{N}$ and $V_f^N(\delta) := \sup_{\substack{|t-s|<\delta \\ s,t \leq N}} |f(t) - f(s)| : \lim_{\delta \to 0} \sup_{f \in A} V_f^N(\delta) = 0$.

So two things can go wrong. Either functions in A are not sufficiently bounded or there are regularity holes/not all functions have the same kind of regularity.

Example 2.3

The function

$$g_n(x) := \begin{cases} x^{1/n} & , x \leq 1 \\ 2 - x & , x \in (1,2) \\ 0 & , x \geq 2 \end{cases}$$

The set $A := \{g_n \mid n \in \mathbb{N}\}$ is neither compact nor relatively compact.

- We need that every sequence has a subsequence that convergence to a $g \in C([0,\infty])$ however,

$$\lim_{n \to \infty} g_n(0) = 0 \text{ and } \forall x \in (0,1] : \quad \lim_{n \to \infty} g_n(x) = 1.$$

  So the limit is discontinuous at zero. So its not compact

- It also fails the second bullet point of Ascela-Ascoli:

$$\forall \epsilon > 0, s \in (0, \epsilon] : \quad \sup_{n \in \mathbb{N}} |g_n(s) - g_n(0)| = 1.$$

We see that this family of functions is for every $n \in \mathbb{N}$ Hölder continuous with index $\gamma = \frac{1}{n}$:

$$|g_n(y) - g_n(x)| \leq |x - y|^{\frac{1}{n}}, \quad \forall x, y \in [0,1].$$

What goes wrong here? Is it the boundedness or regularity? The missing piece is Hölder continuity where the index is independent of $n \in \mathbb{N}$.

Lemma 2.30

Suppose $(K_N)_{N \in \mathbb{N}}$ are non-negative and $\gamma \in (0,1]$, then

$A_K^\gamma := \{f : [0,\infty) \to \mathbb{R} \mid f$ Hölder continuous on compacts with index γ, constant K_N and $|f(0)| \leq K_0\}$

is compact in $C([0,\infty))$.

Theorem 2.31

Suppose $(X^n)_{n\in\mathbb{N}}$ is a sequence of continuous stochastic processes on $(\Omega,\mathcal{F},\mathbb{P})$ such that

- $(\mathbb{P}^{X_0^n})_{n\in\mathbb{N}}$ is tight in $\mathcal{M}_1(\mathbb{R})$
- For every $n\in\mathbb{N}$ and $T>0$ exists c,α,β independent of $n\in\mathbb{N}$ with

$$\forall s,t\le T:\quad \mathbb{E}[|X_t^n-X_s^n|^\alpha]\le C|t-s|^{1+\beta}.$$

Then $(\mathbb{P}^{X^n})_{n\in\mathbb{N}}$ is tight in $\mathcal{M}_1(C([0,\infty)))$.

So previously we had Ascela-Ascoli as a tool to construct compact sets to check for tightness, i.e.

$$\forall\epsilon>0\exists K \text{ compact subsets: } \inf_{n\in\mathbb{N}}\mathbb{P}^{X^n}(K)>1-\epsilon \quad\Longleftrightarrow\quad (\mathbb{P}^{X^n})_{n\in\mathbb{N}} \text{ tight in } \mathcal{M}_1(C([0,\infty))).$$

There we would consider

$$K:=\{f\in C([0,\infty))\mid |f(0)|\le a,\forall N\in\mathbb{N},\delta>0: V_f^N(\delta)\to 0,\delta\to 0\}$$

which leads with the above to an infinite dimensional joint distribution of $(X_t^n)_{t\ge0}$ integral, when computing the tail probability. Kolmogorov Chensov avoids this, by only considering a bi variate expectation.

2.2 Brownian Motion

2.3 Bronwian Family

Theorem 2.32

Let $(\Omega,\mathcal{F},\mathbb{P})$ be a probability space on which $(\xi_i)_{i\in\mathbb{N}}$ is an iid sequence with mean zero and finite variance. Define $X_t^n:=\frac{Y_{nt}}{\sigma\sqrt{n}}$ and $P_n:=\mathbb{P}\circ(X^n)^{-1}$ on $(C([0,\infty)),\mathcal{B}(C([0,\infty)))$. Then $w-\lim_{n\to\infty}P_n=P_*$, where $W\sim P_*$ is the coordinate mapping process on $C([0,\infty))$ is a one-dimensional Brownian.

Definition 2.33

The probability measure P_* on $(C([0,\infty)),\mathcal{B}(C[0,\infty)))$ under which $W_t(\omega)=\omega(t)$, $t\in[0,\infty)$, is a standard one dimensional Brownian motion, called Wiener measure.

Let $P^1,...,P^d$ be d copies of the Wiener measure, then $P^0:=P^1\times\cdots\times P^d$ is a measure on $(C([0,\infty)^d),\mathcal{B}(C([0,\infty)^d)))$ and $B_t(\omega)=\omega(t)$ on P^0 with $(\mathcal{F}_t^B)_{t\ge0}$ is a d-dimensional Brownian motion starting at the origin. Define

$$\forall x\in\mathbb{R}^d, F\in\mathcal{B}(C([0,\infty)^d)):\quad P^x(F):=P^0(F-x),$$

a prbability measure on $(C([0,\infty)^d),\mathcal{B}(C([0,\infty)^d)))$, where

$$F-x:=\{\omega\in C([0,\infty)^d)\mid \omega(\cdot)+x\in F\}.$$

Then under P^x, $(B_t)_{t\ge0}$ is a d dimensional Brownian motion starting in x. For a probability measure μ on $(\mathbb{R}^d,\mathcal{B}(\mathbb{R}^d))$ on $\mathcal{B}(C([0,\infty)^d))$ define

$$P^\mu(F):=\int_{\mathbb{R}^d}P^x(F)d\mu,\qquad \text{on } \mathcal{B}(C([0,\infty)^d)).$$

Lemma 2.34

$\forall F \in \mathcal{B}(C([0,\infty)^d))$ the mapping $x \mapsto P^x F$ is $\mathcal{B}(\mathbb{R}^d)$, $\mathcal{B}([0,1])$ measurable.

Proof. Define $\mathcal{C}$ as the family of all cylinder sets in Ω, constructed by

$$C := \{\omega \in \Omega \mid \omega(t_1) \in B_1, ..., \omega(t_n) \in B_n\}, \quad \forall 0 \leq t_1 < ... < t_n < \infty, B_1, ..., B_n \in \mathcal{B}(\mathbb{R}^d).$$

Then $\mathcal{C}$ is $\cap$-stable and generates $\sigma(\mathcal{C}) = \mathcal{F}$. For, fixed $C \in \mathcal{C}$ $x \mapsto P^x(C)$ is $\mathcal{B}(\mathbb{R}^d)$ measurable, because we can formulate $P^x(C)$ as an integral with continuous densities. Finally, define

$$\mathcal{D} := \{F \in \mathcal{F} \mid x \mapsto P^x(F) \text{ is } \mathcal{B}(\mathbb{R}^d) \text{ measurable}\}.$$

- $\Omega \in \mathcal{D}$, because $x \mapsto P^x(\Omega) = 1$ is a constant and thus measurable
- Let $F \in \mathcal{D}$ then $x \mapsto P^x(F^C) = 1 - P^x(F)$, is just a translation and thus measurable.
- Let $(F_n)_{n\in\mathbb{N}} \subset \mathcal{D}$ then $x \mapsto P^x(\bigcup_{n\in\mathbb{N}} F_n) = \sum_{n=1}^{\infty} P^x(F_n)$ is again measurable.

Thus,

$$\mathcal{F} = \sigma(\mathcal{C}) \subseteq \mathcal{D} \subseteq \mathcal{F} \quad \Longrightarrow \quad \mathcal{D} = \mathcal{F}.$$

□

Definition 2.35

A d-dimensional Brownian family is an adapted, d-dimensional process $(W_t)_{t\geq 0}$ on a filtered measurable space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0})$ and a family of probability measures $(P^x)_{x\in\mathbb{R}^d}$ such that

- $\forall F \in \mathcal{F}$ the mapping $x \mapsto P^x(F)$ is (universally ???) measurable
- $\forall x \in \mathbb{R}^d$: $P^x(W_0 = x) = 1$
- $\forall P^x$ the process W is a d-dimensional Brownian motion starting in x.

2.4 Weak Solutions by Means of the Girsanov Theorem

Proposition 2.36

For $T > 0$ fixed consider

$$dX_t = b(t, X_t)dt + dW_t, \quad t \in [0, T],$$

where W is d-dimensional Brownian motion, $b : [0, T] \times \mathbb{R}^d \to \mathbb{R}^d$, is Borel measurable and satisfies

$$\forall x \in \mathbb{R}^d, t \in [0, T] : \quad \|b(x, t)\| \leq K(1 + \|x\|),$$

for some constant K. Then for any initial distribution μ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ the SDE has a weak solution with initial distribution μ.

Proof. Beginning with a Brownian family $(W_t)_{t\geq 0}$, $(\mathcal{F}_t)_{t\geq 0}$ and $(\mathbb{P}^x)_{x\in\mathbb{R}^d}$ then

$$Z_t := e^{\sum_{j=1}^d \int_0^t b_j(s,X_s)dX_s^j - \frac{1}{2}\int_0^t \|b(s,X_s)\|^2 ds}$$

is a martingale under each $\mathbb{P}^x$. Under Q^x given by $\frac{dQ^x}{d\mathbb{P}^x} =: Z_T$

$$W_t := X_t - X_0 - \int_0^t b(s, X_s)ds, \quad t \in [0, T],$$

is a Brownian motion with $Q^x(W_0 = 0) = 1$, $\forall x \in \mathbb{R}^d$. The above is equivalent to

$$X_t = X_0 - \int_0^t b(s, X_s)ds + W_t, \quad t \in [0, T].$$

With $Q^\mu(\cdot) := \int_{\mathbb{R}^d} Q^x(\cdot)d\mu(x)$, then $((X, W), (\Omega, \mathcal{F}, Q^\mu), (\mathcal{F}_t))$ is a weak solution. □

Remark 2.4

To find a weak solution for $t \in [0, \infty)$, the finite-time Girsanov theorem needs to be extended. The Radon-Nikodym derivative $Z_T := \frac{dQ^x}{d\mathbb{P}^x}$ is only defined on the bounded σ-algebra $\mathcal{F}_T$.
For $M_t^j := \int_0^t b(s, X_s)dX_s^j$ yields that $\int_0^t \|b(s, X_s)\|^2 ds = \langle M \rangle_t$. Then

$$Z_\infty = \lim_{t\to\infty} e^{\sum_{j=1}^d M_t^j - \frac{1}{2}\langle M\rangle_t} \notin L^1(\mathbb{P}).$$

So $(Z_t)_{t\in[0,\infty)}$ is not a martingale. Furthermore, with Fatou

$$\mathbb{E}[\liminf_{t\to\infty} Z_t] \le \liminf_{t\to\infty} \mathbb{E}[Z_t] = 1.$$

This means we cannot find a Girsanov measure Q^x such that $Q^x(A) = \int_A Z_\infty d\mathbb{P}^x$ for any $A \in \mathcal{B}(\mathbb{R}^d)$, because

$$1 = Q^x(\Omega) \neq \int_\Omega Z_\infty d\mathbb{P}^x < 1.$$

So Z_∞ also cannot be Radon derivative.
Let Ω be infinite set and $\mathcal{A}$ an algebra. By Zorn there exists $\mathcal{U} \subset \mathcal{A}$ such that $\forall A, B \in \mathcal{U} : A \cap B \neq \emptyset$ but $\bigcap_{A\in\mathcal{U}} A = \emptyset$. Define

$$Q_p : \mathcal{A} \to \{0, 1\}, \quad A \mapsto Q_p(A) = \begin{cases} 1 & , A \in \mathcal{U} \\ 0 & , A \notin \mathcal{U} \end{cases}.$$

Because $\bigcap_{A\in\mathcal{U}} A = \emptyset$ there exist $A_1 \supset A_2 \supset \ldots$ in $\mathcal{U}$ such that $\bigcap_{n\in\mathbb{N}} A_n = \emptyset$, with $Q_p(A_n) = 1$ for every $n \in \mathbb{N}$. Thus, $\lim_{n\to\infty} Q_p(A_n) = 1 \neq 0$, which is a contradition. So Q_p cannot be extended to $\sigma(A)$.

2.5 Stochastic Calculus Backround

Remark 2.5

We discuss the notion of an extended probability space. Let $(X_t)_{t\ge0}$ be $\mathcal{F}_t$-adapted process on $(\Omega, \mathcal{F}, \mathbb{P})$. If one requires a d-dimensional Brownian motion independent of X, one might run into the issue that the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ might not be rich enough to support this Brownian motion.

Example 2.4

Consider $\Omega := C([0, \infty), \mathbb{R}^d)$ and $\mathcal{F} := \sigma(X_s \mid s \ge 0) = \sigma(X)$, then $X_t(\omega) = \omega(t)$ is a stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$. Any Brownian motion on that probability space satisfies $\sigma(W) \subseteq \mathcal{F} = \sigma(X)$. Doobs factorization lemma yields that for every $t \ge 0$ there exists a $\mathbb{P}_X$-a.s. unique measurable mapping $\varphi_t := C([0, \infty), \mathbb{R}^d) \to \mathbb{R}^d$ such that $W_t = \varphi_t(X)$. So W is just a transformation of X.

Remark 2.6

To extend the probability space to construct a Brownian motion independent of X, let $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathbb{P}})$ be another probability space on which $\hat{W}$ is a Brownian motion and set $\tilde{\Omega} := \Omega \times \hat{\Omega}$, $\tilde{\mathcal{G}} := \mathcal{F} \otimes \hat{\mathcal{F}}$, $\tilde{\mathbb{P}} := \mathbb{P} \otimes \hat{\mathbb{P}}$ and the filtration $\tilde{\mathcal{G}}_t := \mathcal{F}_t \otimes \hat{\mathcal{F}}_t$. We make the filtration right continuous by defining

$$\tilde{\mathcal{F}}_t := \bigcap_{s>t} \sigma(\tilde{\mathcal{G}} \cup \mathcal{N}),$$

where $\mathcal{N}$ are the $\tilde{\mathbb{P}}$-nullsets in $\tilde{\mathcal{G}}$. We make the σ-algebra complete by $\tilde{\mathcal{F}} := \sigma(\tilde{\mathcal{G}} \cup \mathcal{N})$. Finally, we extend X and W to be $\tilde{\mathcal{F}}_t$-adapted processes on $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbb{P}})$ by defining

$$\forall (\omega, \hat{\omega}) \in \tilde{\Omega}: \quad \tilde{X}_t(\omega, \hat{\omega}) := X_t(\omega) \quad \text{and} \quad \tilde{W}_t(\omega, \hat{\omega}) := W_t(\hat{\omega}).$$

Then $(\tilde{W}_t)_{t\geq 0}$ is independent of $(\tilde{X}_t)_{t\geq 0}$.

Theorem 2.37

Let $(M_t := (M_t^1, ..., M_t^d))_{t\geq 0}$ defined on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ with M^i being a continuous local martingale for every $i = 1, ..., d$. Also suppose that the cross-variation $\langle M^i, M^j \rangle_t$ is $\mathbb{P}$-a.s. an absolutely continuous function in t for every $i, j = 1, ..., d$. Then there exists an extension $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbb{P}})$ of $(\Omega, \mathcal{F}, \mathbb{P})$ on which we define a Brownian motion W and $((X_t^{i,k})_i, k = 1, ..., d)_{t\geq 0}$ a metric of measurable adapted processes with

$$\forall t \in [0, \infty), i, k = 1, ..., d : \tilde{\mathbb{P}}(\int_0^t |X_s^{i,k}|^2 ds < \infty) = 1,$$

such that

$$M_t^i = \sum_{k=1}^d \int_0^t X_s^{i,k} dW_s^k \quad i = 1, ..., d, t \in [0, \infty) \quad \tilde{\mathbb{P}} - a.s.$$

and

$$\langle M^i, M^j \rangle_t = \sum_{k=1}^d \int_0^t X_s^{i,k} X_s^{j,k} ds, \quad i, j = 1, ..., d, t \in [0, \infty) \quad \tilde{\mathbb{P}} - a.s.$$

Proof. We reduce the d-dimensional process by a random, time dependent rotation of coordinates to one-dimensional processes. Define

$$z_t^{i,j} := \frac{d}{dt} \langle M^i, M^j \rangle_t = \lim_{n\to\infty} n(\langle M^i, M^j \rangle_t - \langle M^i, M^j \rangle_{(t-\frac{1}{n})^+}).$$

As $z_t^{i,j} = z_t^{j,i}$, because $\langle cM^i \rangle_t = c^2 \langle M^i \rangle_t$ yields

$$\langle M^i, M^j \rangle_t = \frac{1}{4}(\langle M^i + M^j \rangle_t - \langle M^i - M^j \rangle_t) = \langle M^j, M^i \rangle_t,$$

we get that the matrix process $Z := (Z_t := (z_t^{ij})_{i,j=1,...,d})_{t\geq 0}$ is symmetric and $(s, \omega) \mapsto Z_s(\omega)$ is $\mathcal{B}([0, t])???\otimes$

$\mathcal{F}_t$-measurable, i.e. progressively measurable. Additionally, we have

$$\begin{aligned}\forall \alpha \in \mathbb{R}^d : \ \alpha^T Z_t \alpha &= \lim_{n\to\infty} n(\sum_{i=1}^d \sum_{j=1}^d \alpha_i(\langle M^i, M^j\rangle_t - \langle M^i, M^j\rangle_{(t-\frac{1}{n})^+})\alpha_j) \\ &= \lim_{n\to\infty} n(\langle \sum_{i=1}^d \alpha_i M^i\rangle_t - \langle \sum_{i=1}^d \alpha_i M^j\rangle_{(t-\frac{1}{n})^+}) \\ &= \frac{d}{dt}\langle \sum_{i=1}^d \alpha_i M^i\rangle_t \geq 0.\end{aligned}$$

Because the quadratic variation is non-decreasing, for fixed $\omega \in \Omega$ it is $\lambda - a.e.$ differentiable in $t \geq 0$ with Lebesgues theorem.
Because Z is progressively measurable the set

$$E := \{(t,\omega) \in [0,\infty) \times \Omega \mid Z_t(\omega) \text{ is not positive semi definite}\} \in \mathcal{B}([0,\infty)) \otimes \mathcal{F}.$$

This yields

$$(\lambda \otimes \mathbb{P})(E) = \int_\Omega \lambda(\{t \geq 0 \mid (t,\omega) \in E\}) d\mathbb{P}(\omega) = \int_\Omega 0 d\mathbb{P}(\omega) = 0.$$

So Z_t is positive semi definite for λ-a.e $t \geq 0$ $\mathbb{P}$-a.s.
It is well known that any symmetric, positive definite matrix can be diagonalized by an orthogonal matrix, i.e. there exists Q such that

$$Q^{-1} Z Q = \Lambda,$$

where Λ is the daigonal matrix with the non negative eigenvalues of Z. So there exists progressively measurable matrix valued processes

$$\left(Q_t := (q_t^{i,j})_{i,j=1,\dots,d}\right)_{t\geq 0} \quad \text{and} \quad \left(\Lambda_t := (\delta_{ij}\lambda_t^i)_{i,j=1,\dots,d}\right)_{t\geq 0},$$

such that for λ-a.e. $t \geq 0$ it holds

$$\sum_{k=1}^d q_t^{k,i} q_t^{k,j} = (q_t^{\cdot,i})^T q_t^{\cdot,j} = (q_t^{i,\cdot})^T q_t^{j,\cdot} = \sum_{k=1}^d q_t^{i,k} q_t^{j,k} = \delta_{ij}, \ i,j = 1,\dots,d \quad \mathbb{P} - a.s.$$

and

$$(q^{\cdot,i})^T Z_t (q^{\cdot,j}) = \sum_{k=1}^d \sum_{l=1}^d q_t^{k,i} z_t^{k,l} q_t^{l,j} = \delta_{ij}\lambda_t^i \geq 0, i,j = 1,\dots,d \quad \mathbb{P} - a.s.$$

From the orthogonality, we see that $(q_t^{k,i})^2 \leq 1$ and thus

$$\int_0^t (q_s^{k,i})^2 d\langle M^k\rangle_s \leq \langle M^k\rangle_t < \infty.$$

Then, $N_t^i := \sum_{k=1}^d \int_0^t q_s^{k,i} dM_s^k, \quad i = 1,\dots,d, t \in [0,\infty)$ is a continuous local martingale.
To finish the proof our goal is to represent M^i in terms of an independent Brownian motion on an extended probability space $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbb{P}})$. The next step is to represent N^i in terms of that independent Brownian motion. First, see that

$$\begin{aligned}\langle N^i, N^j\rangle_t &= \sum_{k=1}^d \sum_{l=1}^d \int_0^t q_s^{k,i} q_s^{l,j} d\langle M^k, M^l\rangle_s \\ &= \sum_{k=1}^d \sum_{l=1}^d \int_0^t q_s^{k,i} q_s^{l,j} \frac{d}{dt}\langle M^k, M^l\rangle_s ds \\ &= \sum_{k=1}^d \sum_{l=1}^d \int_0^t q_s^{k,i} z_s^{k,l} q_s^{l,j} ds = \delta_{ij} \int_0^t \lambda_s^i ds\end{aligned}$$

Because N^i is a continuous local martingale $\langle N^i\rangle_t < \infty$, which implies that the Lebesgues-Stieltjes integral $\int_0^t \mathbf{1}_{\lambda_s^i>0}\frac{1}{\lambda_s^i}d\langle N^i\rangle_s$ exists. As

$$\langle N^i\rangle_t = \langle N^i, N^i\rangle_t = \int_0^t \lambda_s^i ds,$$

it follows $\frac{d}{dt}\langle N^i\rangle_t = \lambda_s^i$, which yields

$$\int_0^t \mathbf{1}_{\lambda_s^i>0}\frac{1}{\lambda_s^i}d\langle N^i\rangle_s = \int_0^t \mathbf{1}_{\lambda_s^i>0}ds \leq t.$$

This gives us that

$$W_t^i := \int_0^t \mathbf{1}_{\lambda_s^i>0}\frac{1}{\sqrt{\lambda_s^i}}dN_s^i + \int_0^t \mathbf{1}_{\lambda_s^i=0}d\tilde{W}_s^i, \quad i = 1, ..., d,$$

is a continuous local martingale, where $\tilde{W}$ is a Brownian motion on the extended probability space $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbb{P}})$. Because

$$\forall t \in [0,\infty), i, j = 1, ..., d: \quad \langle W^i, W^j\rangle_t = \delta_{ij}t,$$

we have that W is a Brownian motion wrt. $(\tilde{\mathcal{F}}_t)_{t\geq 0}$. Moreover,

$$\int_0^t \sqrt{\lambda_s^i}dW_s^i = \int_0^t \mathbf{1}_{\lambda_s^i>0}dN_s^i + \int_0^t \sqrt{\lambda_s^i}\mathbf{1}_{\lambda_s^i=0}d\tilde{W}_s^i = \int_0^t \mathbf{1}_{\lambda_s^i>0}dN_s^i \overset{(*)}{=} \int_0^t \mathbf{1}_{\lambda_s^i\geq 0}dN_s^i = N_t^i, \quad \tilde{\mathbb{P}} - a.s.$$

although we used in (*) that the martingale $\int_0^t \mathbf{1}_{\lambda_s^i=0}dN_s^i$ has quadratic variation

$$\langle \int_0^\cdot \mathbf{1}_{\lambda_s^i=0}dN_s^i\rangle_t = \int_0^t \mathbf{1}_{\lambda_s^i=0}d\langle N^i\rangle_s = \int_0^t \mathbf{1}_{\lambda_s^i=0}\lambda_s^i ds = 0$$

and starting in zero yields that it is the trivial martingale, i.e.

$$\int_0^t \mathbf{1}_{\lambda_s^i=0}dN_s^i = 0 \quad , t \geq 0 \quad \tilde{\mathbb{P}} - a.s.$$

Setting $X_t^{i,k} := q_t^{i,k}\sqrt{\lambda_t^k}$ we get that

$$\begin{aligned}\forall t \geq 0, i, k = 1, ..., d: \quad \tilde{P}(\int_0^t (X_s^{i,k})^2 ds < \infty) &= \tilde{P}(\int_0^t (q_s^{i,k})^2\lambda_s^k ds < \infty)\\ &= \tilde{P}(\int_0^t \lambda_s^k ds < \infty) = \tilde{P}(\langle N^k\rangle_t < \infty) = 1.\end{aligned}$$

Finally,

$$\begin{aligned}\sum_{k=1}^d \int_0^t X^{i,k}dW_s^k &= \sum_{k=1}^d \int_0^t q_s^{i,k}\sqrt{\lambda_s^k}dW_s^k = \sum_{k=1}^d \int_0^t q_s^{i,k}dN_s^k\\ &= \sum_{j=1}^d\sum_{k=1}^d \int_0^t q_s^{i,k}q_s^{j,k}dM_s^j \overset{orthogonal}{=} \sum_{j=1}^d \delta_{ij}\int_0^t dM_s^j = M_t^i.\end{aligned}$$

The last equation follows directly from this one. □

Remark 2.7

Note that if the matric process $Z_t := (z_t^{i,j})_{i,j=1,\dots,d}$ has constant rank r $\mathbb{P}$-a.s. λ-almoust every $t \geq 0$, then

$$\forall i \leq r: \quad W_t^i = \int_0^t \mathbf{1}_{\lambda_s^i > 0} \frac{1}{\sqrt{\lambda_s^i}} dN_s^i,$$

is a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ and there is no need to extend the probability space, as it is independently defined from $\tilde{W}$.

SECTION 3

GIRSANOV THEOREM

We study local martingales if a different equivalent probability measure Q on $(\Omega, \mathcal{F})$ is chosen.

Definition 3.1

We call two probability measures Q and $\mathbb{P}$ on $(\Omega, \mathcal{F})$ equivalent if

$$\forall A \in \mathcal{F}: \quad \mathbb{P}(A) = 0 \quad \Longleftrightarrow \quad Q(A) = 0.$$

By Radon-Nikodym there exists a Radon density L_T on $(\Omega, \mathcal{F})$ with $L_T := \frac{dQ}{d\mathbb{P}}$ such that

$$\forall A \in \mathcal{F}: \quad \mathbb{E}^Q[\mathbf{1}_A] = \mathbb{E}[\mathbf{1}_A L_T].$$

If $L_T \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ define $L_t := \mathbb{E}[L_T \mid \mathcal{F}_t]$ for every $t \in [0, T]$, then $(L_t)_{t\in[0,T]}$ is a martingale (???) and is called density process of Q wrt. $\mathbb{P}$.

Assumption 3.2

The stochastic process $(L_t)_{t\in[0,T]}$ satisfies:

- $(L_t)_{t\in[0,T]}$ is a continuous $\mathbb{P}$-martingale
- $\forall t \in [0, T]$: $L_t > 0$ $\mathbb{P}$-a.s. and $L_0 = 1$.

Under these assumptions we can define the associated probability measure

$$Q(A) := \mathbb{E}^Q[\mathbf{1}_A] = \mathbb{E}[\mathbf{1}_A L_T], \quad \forall A \in \mathcal{F}.$$

Then Q is equivalent to $\mathbb{P}$ and with Radon-Nikodym the density is $\frac{dQ}{d\mathbb{P}} = L_T$.

Proposition 3.3

Let $(L_t)_{t\in[0,T]}$ satisfy the assumptions above, where Q is the associated probability measure and $(X_t)_{t\in[0,T]}$ an adapted, continuous stochastic process, then

$$(L_t X_t)_{t\in[0,T]} \text{ is a (local) } \mathbb{P}\text{-martingale} \quad \Longleftrightarrow \quad (X_t)_{t\in[0,T]} \text{ is a (local) } Q\text{-martingale.}$$

Proof. "$\Rightarrow$"
Because X is adapted and Q integrable, it holds

$$\mathbb{E}^Q[|X_t|] = \mathbb{E}[L_T|X_t|] = \mathbb{E}[\mathbb{E}[L_T|X_t| \mid \mathcal{F}_t]] = \mathbb{E}[L_t|X_t|] = \mathbb{E}[|L_tX_t|] < \infty,$$

because L and LX are $\mathbb{P}$-martingales. In order to show the martingale property it suffices to show:

$$\forall A \in \mathcal{F}_s : \quad \mathbb{E}^Q[\mathbf{1}_A(X_t - X_s)] = 0.$$

This immediately follows from

$$\begin{aligned}\mathbb{E}^Q[\mathbf{1}_A(X_t - X_s)] &= \mathbb{E}[L_t\mathbf{1}_A(X_t - X_s)] \\ &= \mathbb{E}[\mathbb{E}[L_T\mathbf{1}_AX_t \mid \mathcal{F}_t] - \mathbb{E}[L_T\mathbf{1}_AX_s \mid \mathcal{F}_s]] \\ &= \mathbb{E}[L_t\mathbf{1}_AX_t - L_s\mathbf{1}_AX_s] = \mathbb{E}[\mathbf{1}_A(L_tX_t - L_sX_s)] \overset{martingale}{=} 0.\end{aligned}$$

"$\Leftarrow$"
Same arguments. □

Theorem 3.4

If

$$L_t := e^{\int_0^t X_s dB_s - \frac{1}{2}\int_0^t X_s^2 ds}, \quad t \in [0,T]$$

and $(L_t)_{t\in[0,T]}$ is martingale on $(\Omega, \mathcal{F}_T, \mathbb{P})$ wrt. the Brownian standard filtration $(\mathcal{F}_t)_{t\in[0,T]}$, then

$$\tilde{B}_t := B_t - \int_0^t X_s ds, \quad t \in [0,T],$$

is Brownian motion wrt. $(\Omega, \mathcal{F}_T, (\mathcal{F}_t)_{t\in[0,T]}, Q)$ where $\frac{dQ}{d\mathbb{P}} = L_T$.

Proof. We need to show that $\tilde{B}$ is continuous local Q-martingale with $\langle \tilde{B} \rangle_t = t$, $t \in [0,T]$.

- To show: $\tilde{B}$ is continuous Q-martingale.
  With proposition before we need to show $(\tilde{B}_tL_t)_{t\in[0,T]}$ is continuous local $\mathbb{P}$-martingale. We can apply Ito Formula for L_t, because we can write it as

  $$L_t = f(U_t, U_t) := e^{U_t + Y_t}, \quad U_t := \int_0^t X_s dB_s, \quad V_t := -\frac{1}{2}\int_0^t X_s^2 ds.$$

  Because integrating over $\langle U \rangle_\cdot$ and $\langle V \rangle_\cdot$ yields a Lebesgues Stieltjes integral which leads to $X_s^2 ds$ and 0 respectively. $d\langle U, V \rangle_s = X_s^2 \cdot 0 ds = 0$. $f(0,0) = 1$ and Itos formula yields

  $$\begin{aligned}L_t &= 1 + \int_0^t L_s dU_s + \int_0^t L_s dV_s + \frac{1}{2}\int_0^t L_s X_s^2 ds \\ &= 1 + \int_0^t L_s X_s dB_s - \frac{1}{2}\int_0^t L_s X_s^2 ds + \frac{1}{2}\int_0^t L_s X_s^2 ds \\ &= 1 + \int_0^t L_s X_s dB_s\end{aligned}$$

Thus, $dL_t = L_s X_s dB_s$, which yields with the product rule

$$\begin{aligned}\tilde{B}_t L_t &= \tilde{B}_0 L_0 + \int_0^t \tilde{B}_s dL_s + \int_0^t L_s d\tilde{B}_s + \langle B, L\rangle_t \\ &= 0 + \int_0^t \tilde{B}_s L_s X_s dB_s + \int_0^t L_s dB_s - \underbrace{\int_0^t L_s X_s ds + \int_0^t 1 \cdot L_s X_s ds}_{=0} \\ &= \int_0^t (\tilde{B}_s X_s + 1) L_s dB_s,\end{aligned}$$

which is a continuous local martingale wrt. $\mathbb{P}$.

- $(\tilde{B}_t)_{t\in[0,T]}$ is a continuous local Q-martingale.
  Let $((\tilde{B}L)_{t\wedge\tau_n})_{t\in[0,T]}$ be a $\mathbb{P}$-martingale for some suitable localizing stopping times $\tau_n \nearrow T$. With the first step and the proposition from before $(\tilde{B}_{t\wedge\tau_n})_{t\in[0,T]}$ is a continuous Q-martingale wrt. $(\mathcal{F}_t)_{t\in[0,T]}$ it follows that $(\tilde{B}_t)_{t\in[0,T]}$ is a continuous local Q-martingale.

- $\forall t \in [0,T]$: $\langle \tilde{B}\rangle_t = t$ under Q.
  To this end, we need to show that $(\tilde{B}_t - t)_{t\in[0,T]}$ is a continuous local martingale under Q. With the proposition from before it suffices to show that $((\tilde{B}_t^2 - t)L_t)_{t\in[0,T]}$ is a continuous local $\mathbb{P}$-martingale. Then, because the quadratic variation of any finite variation process, e.g. $\int_0^\cdot X_s ds$, is zero and and its co-variation with a continuous martingale is zero, under $\mathbb{P}$ we have

$$\langle \tilde{B}\rangle_t = \langle B - \int_0^\cdot X_s ds\rangle_t = \langle B\rangle_t - 2\langle B, \int_0^\cdot X ds\rangle_t + \langle \int_0^\cdot X ds\rangle_t = \langle B\rangle_t - 0 + 0 = t.$$

  Finally, with Itos formula for Ito processes for $f(x) = x^2$ we have

$$\tilde{B}_t^2 = \int_0^t 2\tilde{B}_s d\tilde{B}_s + \int_0^t \langle \tilde{B}\rangle_s = \int_0^t 2\tilde{B}_s d\tilde{B}_s + t \overset{Def.}{=} \int_0^t 2\tilde{B}_s dB_s - \int_0^t 2\tilde{B}_s X_s ds + t.$$

  This yields with the product formula:

$$\begin{aligned}(\tilde{B}_t^2 - t)L_t &= \int_0^t (B_s^2 - s) dL_s + \int_0^t L_s d(\tilde{B}_s - s) + \langle \tilde{B}^2 - \cdot, L\rangle_t \\ &= \int_0^t (B_s^2 - s) dL_s + \int_0^t L_s 2\tilde{B}_s d\tilde{B}_s + \int_0^t 2\tilde{B}_s L_s X_s ds \\ &= \int_0^t (B_s^2 - s) L_s X_s dB_s + \int_0^t 2L_s \tilde{B}_s dB_s - \int_0^t 2L_s \tilde{B}_s X_s ds + \int_0^t 2\tilde{B}_s L_s X_s ds \\ &= \int_0^t ((\tilde{B}_s - s)X_s + 2\tilde{B}_s) L_s dB_s,\end{aligned}$$

  which is a continuous local $\mathbb{P}$-martingale.

□

SECTION 4

WEAK SOLUTIONS

The Brownian motion is now denoted by (W_t).
One can prove the existence of a solution in a weaker sense for less restrictive assumptions on the drift term

Definition 4.1

Let $a : [0,\infty) \times \mathbb{R}^d \to \mathbb{R}^d$ and $b : [0,\infty) \times \mathbb{R}^d \to \mathbb{R}^{d\times r}$ be Borel measurable functions. For a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$, a r-dimensional $(\mathcal{F}_t)$-Brownian motion W and a $\mathcal{F}_0$ measurable random variable ξ, we call call $(X_t)_{t\geq 0}$ strong solution to the SDE

$$dX_t = a(t, X_t)dt + b(t, X_t)dW_t,$$

if

- $(X_t)_{t\geq 0}$ is $(\mathcal{F}_t^W \vee \sigma(\xi) \vee \mathcal{N})_{t\geq 0}$ adapted to the augmented filtration
- $$\forall i = 1, ..., d, j = 1, ..., r \text{ and } t \in [0,\infty) : \quad \mathbb{P}(\int_0^t |a_i(s, X_s)| + b_{ij}^2(s, X_s)ds < \infty) = 1$$
- $$\forall t \in [0, T], i = 1, ..., d : \quad X_t^i = X_0^i + \int_0^t a_i(s, X_s)ds + \sum_{j=1}^r \int_0^t b_{ij}(s, X_s)dW_s^j.$$

Definition 4.2

A weak solution, with initial distribution μ on $\mathcal{B}(\mathbb{R}^d)$ of

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t$$

is a tuple $((\Omega, \mathcal{F}, \mathbb{P}), (\mathcal{F}_t)_{t\geq 0}, X, W)$ where $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ is a filtered probability space, where the filtration satisfies the usual conditions, X is $(\mathcal{F}_t)$-adapted, $\mathbb{R}^d$ valued process and W a r-dimensional $(\mathcal{F}_t)$-Brownian motion W

- $$\forall i = 1, ..., d, j = 1, ..., r \text{ and } t \in [0,\infty) : \quad \mathbb{P}(\int_0^t |a_i(s, X_s)| + b_{ij}^2(s, X_s)ds < \infty) = 1$$
- $$\forall t \in [0, T], i = 1, ..., d : \quad X_t^i = X_0^i + \int_0^t a_i(s, X_s)ds + \sum_{j=1}^r \int_0^t b_{ij}(s, X_s)dW_s^j.$$

Remark 4.1

For a strong solution the probability space was given. For weak solutions, the probability space is provided as part of the solution. Further, the randomness in a strong solution only comes from the randomness of the augmented standard Brownian filtration, whereas the randomness in a weak solution comes from any filtration, with the restriction that W has to be a $(\mathcal{F}_t)$-Brownian motion. This means, that the solution at time t can not only be given by the initial condition, a measurable functional given by the Brownian path and the drift, but also from randomness independent from the initial condition and the Brownian motion.

Example 4.1

Consider

$$dX_t = b(t, X_t)dt + dW.t, \quad t \in [0, T], \quad T < \infty$$

and $b : \mathbb{R}^d \to \mathbb{R}^d$ is a measurable function with sub-linear growth

$$\forall x \in \mathbb{R}^d, t \in [0, T] : \quad \|b(t, x)\| \leq K(1 + \|x\|).$$

Let X be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ and let $(\mathcal{F}_t)_{t\geq 0}$ be the augmented filtration generated by X, i.e.

$$\forall t \geq 0 : \quad \mathcal{F}_t := \sigma(\mathcal{F}_t^X \cup \mathcal{N}), \quad \mathcal{N} := \{A \subseteq \Omega \mid \exists C \in \mathcal{F}, \mathbb{P}(C) = 0 : \quad A \subseteq C\}.$$

Define $\forall t \in [0, T]$:

$$Z_t := e^{\int_0^t b(s,X_s)dX_s - \frac{1}{2}\int_0^t \|b(s,X_s)\|^2 ds},$$

which is a martingale under $\mathbb{P}$. Define Q via the Radon-Nikodym derivative

$$\frac{dQ}{d\mathbb{P}} = Z_T, \quad Q(\cdot) := \mathbb{E}^Q[\mathbf{1}.].$$

With Girsanov, the process

$$\tilde{W}_t := X_t - X_0 - \int_0^t b(s, X_s)ds, \quad t \in [0, T]$$

is Brownian motion with $Q(\tilde{W}_0 = 0) = 1$. Then $(X, \tilde{W}_t)$ and $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, Q)$ is weak solution.

4.1 Weak Solutions by Means of the Girsanov Theorem

Proposition 4.3

For $T > 0$ fixed consider

$$dX_t = b(t, X_t)dt + dW_t, \quad t \in [0, T],$$

where W is d-dimensional Brownian motion, $b : [0, T] \times \mathbb{R}^d \to \mathbb{R}^d$, is Borel measurable and satisfies

$$\forall x \in \mathbb{R}^d, t \in [0, T] : \quad \|b(x, t)\| \leq K(1 + \|x\|),$$

for some constant K. Then for any initial distribution μ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ the SDE has a weak solution with initial distribution μ.

Proof. Beginning with a Brownian family $(W_t)_{t\geq 0}$, $(\mathcal{F}_t)_{t\geq 0}$ and $(\mathbb{P}^x)_{x\in\mathbb{R}^d}$ then

$$Z_t := e^{\sum_{j=1}^d \int_0^t b_j(s,X_s)dX_s^j - \frac{1}{2}\int_0^t \|b(s,X_s)\|^2 ds}$$

is a martingale under each $\mathbb{P}^x$. Under Q^x given by $\frac{dQ^x}{d\mathbb{P}^x} =: Z_T$

$$W_t := X_t - X_0 - \int_0^t b(s, X_s)ds, \quad t \in [0, T],$$

is a Brownian motion with $Q^x(W_0 = 0) = 1$, $\forall x \in \mathbb{R}^d$. The above is equivalent to

$$X_t = X_0 - \int_0^t b(s, X_s)ds + W_t, \quad t \in [0, T].$$

With $Q^\mu(\cdot) := \int_{\mathbb{R}^d} Q^x(\cdot) d\mu(x)$, then $((X,W),(\Omega,\mathcal{F},Q^\mu),(\mathcal{F}_t))$ is a weak solution. □

Remark 4.2

To find a weak solution for $t \in [0,\infty)$, the finite-time Girsanov theorem needs to be extended. The Radon-Nikodym derivative $Z_T := \frac{dQ^x}{d\mathbb{P}^x}$ is only defined on the bounded σ-algebra $\mathcal{F}_T$.
For $M_t^j := \int_0^t b(s,X_s) dX_s^j$ yields that $\int_0^t \|b(s,X_s)\|^2 ds = \langle M \rangle_t$. Then

$$Z_\infty = \lim_{t\to\infty} e^{\sum_{j=1}^d M_t^j - \frac{1}{2}\langle M\rangle_t} \notin L^1(\mathbb{P}).$$

So $(Z_t)_{t\in[0,\infty)}$ is not a martingale. Furthermore, with Fatou

$$\mathbb{E}[\liminf_{t\to\infty} Z_t] \le \liminf_{t\to\infty} \mathbb{E}[Z_t] = 1.$$

This means we cannot find a Girsanov measure Q^x such that $Q^x(A) = \int_A Z_\infty d\mathbb{P}^x$ for any $A \in \mathcal{B}(\mathbb{R}^d)$, because

$$1 = Q^x(\Omega) \neq \int_\Omega Z_\infty d\mathbb{P}^x < 1.$$

So Z_∞ also cannot be Radon derivative.
Let Ω be infinite set and $\mathcal{A}$ an algebra. By Zorn there exists $\mathcal{U} \subset \mathcal{A}$ such that $\forall A,B \in \mathcal{U} : A \cap B \neq \emptyset$ but $\bigcap_{A\in\mathcal{U}} A = \emptyset$. Define

$$Q_p : \mathcal{A} \to \{0,1\}, \quad A \mapsto Q_p(A) = \begin{cases} 1 & , A \in \mathcal{U} \\ 0 & , A \notin \mathcal{U} \end{cases}.$$

Because $\bigcap_{A\in\mathcal{U}} A = \emptyset$ there exist $A_1 \supset A_2 \supset \ldots$ in $\mathcal{U}$ such that $\bigcap_{n\in\mathbb{N}} A_n = \emptyset$, with $Q_p(A_n) = 1$ for every $n \in \mathbb{N}$. Thus, $\lim_{n\to\infty} Q_p(A_n) = 1 \neq 0$, which is a contradition. So Q_p cannot be extended to $\sigma(A)$.

SECTION 5

THE MARTINGALE PROBLEM OF STROOCK AND VARADHAN

Weak solutions is finding a probability space, a Brownian motion and a process with the right law (determined by its finite- dimensional distributions).

Definition 5.1

We say a weak solution is unique in sense of probability law if for any two weak solutions $((X,W),(\Omega,\mathcal{F},\mathbb{P}),(\mathcal{F}_t)_{t\ge0})$ and $((\tilde{X},\tilde{W}),(\tilde{\Omega},\tilde{\mathcal{F}},\tilde{\mathbb{P}}),(\tilde{\mathcal{F}}_t)_{t\ge0})$ with

$$\mathbb{P}(X_0 \in A) = \tilde{\mathbb{P}}(\tilde{X}_0 \in A), \quad A \in \mathcal{B}(\mathbb{R}^d),$$

holds that X and $\tilde{X}$ have the same law, i.e.

$$\mathbb{P} \circ X^{-1}(A) = \tilde{\mathbb{P}} \circ \tilde{X}^{-1}(A), \quad \forall A \in \mathcal{B}(C([0,\infty),\mathbb{R}^d)).$$

There is also a different approach: Search for the law of a diffions process with drift and dispersion coefficients in terms of a martingale problem, which will be equivalent to solving the related SDE in a weak sense. This formulation is useful in terms of continuity and weak convergence arumgents that yield exitnece results and "invariance principles" (i.e. the convergence of Markov chains to diffion processes).

5.1 Fundamentals and Motiviation:

Assume $((X,W),(\Omega,\mathcal{F},\mathbb{P}),(\mathcal{F}_t)_{t\geq 0})$ is a weak solution to the SDE.

Definition 5.2

We call for every u in some function space

$$L : u \mapsto F(x, D^{k_1}, ..., D^{k_s}), \quad \text{where} \quad F : E \times \mathbb{R}^s \to \mathbb{R},$$

although $D^{k_i} := (\partial^{k_1} \cdots \partial^{k_i})$, differential operator on the domain Ω of order s.

For every $t \geq 0$ we introduce the second order differential operator for any $f \in C^2(\mathbb{R}^d)$:

$$(A_t f)(\cdot) := \frac{1}{2}\sum_{i=1}^{d}\sum_{k=1}^{d} a_{ik}(t,\cdot)\frac{\partial^2}{\partial x_i \partial x_k} f(\cdot) + \sum_{i=1}^{d} b_i(t,\cdot)\frac{\partial}{\partial x_i} f(\cdot), \quad \forall f \in C^2(\mathbb{R}^d).$$

Proposition 5.3

Let $(X_t)_{t\geq 0}$ a continuous Ito diffusion on $(\Omega,\mathcal{F},(\mathcal{F}_t)_{t\geq 0},\mathbb{P})$ with generator A_t and diffusion matric $a(t,x) = \sigma(t,x)^T\sigma(t,x)$. For every $f \in C^{1,2}([0,\infty)\times\mathbb{R}^d,\mathbb{R})$ the process $(M_t^f)_{t\geq 0}$ defined by

$$M_t^f := f(t,X_t) - f(0,X_0) - \int_0^t (\frac{\partial}{\partial s} + A_s) f(s,X_s) ds,$$

a continuous local martingale. If g is another Member of $C^{1,2}([0,\infty)\times\mathbb{R}^d,\mathbb{R})$ then M^g is a continuous local martingale and

$$\langle M^f, M^g\rangle_t = \sum_{i=1}^{d}\sum_{k=1}^{d}\int_0^t a_{ik}(s,X_s)\frac{\partial}{\partial x_i} f(s,X_s)\frac{\partial}{\partial x_k} g(s,X_s) ds.$$

Finally, if $f \in C_0([0,\infty)\times\mathbb{R}^d,\mathbb{R})$ (chat sagt $C_c^{1,2}$???) and σ_{ij} are bounded on $suppf$ then M_2^f is a continuous square integrable martingale.

Proof. Using Itos formula for the Ito diffusion

$$dX_t^i = b_i(t,X_t)dt + \sum_{i=1}^{r}\sigma_{ij}(t,X_t)dW_t^j$$

and reminding that

$$d\langle X^i, X^k\rangle_t \overset{???}{=} \sum_{j=1}^{r}\sigma_{ij}(t,X_t)\sigma_{kj}(t,X_t)dt =: a_{ik}(t,X_t)dt$$

yields that

$$\begin{aligned}
df(t,X_t) &= \frac{\partial}{\partial t}f(t,X_t)dt + \sum_{i=1}^{d}\frac{\partial}{\partial x_i}f(t,X_t)\left(b_i(t,X_t)dt + \sum_{j=1}^{r}\sigma_{ij}(t,X_t)dW_t^j\right)\\
&\quad + \frac{1}{2}\sum_{i,k=1}^{d}\frac{\partial^2}{\partial x_i\partial x_k}f(t,X_t)a_{ik}(t,X_t)dt\\
&= \left(\frac{\partial}{\partial t}f(t,X_t) + \sum_{i=1}^{d}b_i(t,X_t)\frac{\partial}{\partial x_i}f(t,X_t) + \frac{1}{2}\sum_{i,k=1}^{d}a_{ik}(t,X_t)\frac{\partial^2}{\partial x_i\partial x_k}f(t,X_t)\right)dt\\
&\quad + \sum_{i=1}^{d}\frac{\partial}{\partial x_i}f(t,X_t)\sum_{j=1}^{r}\sigma_{ij}(t,X_t)dW_t^j\\
&= (\frac{\partial}{\partial t} + \mathcal{A}_t)f(t,X_t)dt + \sum_{i=1}^{d}\sum_{j=1}^{r}\frac{\partial}{\partial x_i}f(t,X_t)\sigma_{ij}(t,X_t)dW_t^j\\
&\iff f(t,X_t) - f(0,X_0) - \int_0^t(\frac{\partial}{\partial s} + \mathcal{A}_s)f(s,X_s)ds = \sum_{i=1}^{d}\sum_{j=1}^{r}\int_0^t\sigma_{ij}(s,X_s)\frac{\partial}{\partial x_i}f(s,X_s)dW_s^j.
\end{aligned}$$

Define the stopping time

$$T_n := \inf\{t\geq 0 \mid \|X\|_t \geq n \vee \exists(i,j): \quad \int_0^t \sigma_{ij}^2(s,X_s)ds \geq n\}.$$

By definition of the weak solution we have

$$\mathbb{P}(\int_0^t \sigma_{ij}^2(s,X_s)ds < \infty) \geq \mathbb{P}(\int_0^t |a_i(s,X_s)| + \sigma_{ij}^2(s,X_s)ds < \infty) = 1,$$

yields $\lim_{n\to\infty} T_n = \infty$ $\mathbb{P}-a.s.$ Then the process defined through $M_t^{f,n} := M_{t\wedge T_n}^f$ is a continuous martingale, because a stochastic intefral $I_t = \int_0^t H_s dW_s$ is a continuous martignale if $\mathbb{E}[\int_0^t H_s^2 ds] < \infty$. This is the case! So M_t^f is a continuous local martingale. The cross variation is

$$\begin{aligned}
\langle M^f, M^g\rangle_t &= \sum_{i=1}^{d}\sum_{j=1}^{r}\langle\int_0^{\cdot\wedge S_n}\sigma_{ij}(s,X_s)\frac{\partial}{\partial x_i}f(s,X_s)dW_s^j, \int_0^{\cdot\wedge S_n}\sigma_{ij}(s,X_s)\frac{\partial}{\partial x_i}g(s,X_s)dW_s^j\rangle_t\\
&\overset{???}{=} \sum_{i=1}^{d}\sum_{k=1}^{d}\int_0^t\sigma_{ij}^2(s,X_s)\frac{\partial}{\partial x_i}f(s,X_s)\frac{\partial}{\partial x_k}g(s,X_s)ds.
\end{aligned}$$

Finally, if f has compact support on which σ_{ij} is bounded, then the integrand $\sigma_{ij}(s,X_s)\frac{\partial}{\partial x_i}f(s,X_s)$ is bounded. So M^f is a continuous and square integrable martingale. □

Remark 5.1

Problem:
Suppose that $((X,W),(\Omega,\mathcal{F},\mathbb{P}),(\mathcal{F}_t)_{t\geq 0})$ is a weak solution to the SDE on $C([0,\infty)^d)$. Set

$$(\mathcal{A}_t'u)(y) := \frac{1}{2}\sum_{i=1}^{d}\sum_{k=1}^{d}a_{ik}(t,y)\frac{\partial^2}{\partial x_i\partial x_k}u(y(t)) + \sum_{i=1}^{d}b_i(t,y)\frac{\partial}{\partial x_i}u(y(t)), t\in[0,\infty), y\in C([0,\infty)^d), u\in C^2(\mathbb{R}^d).$$

For any $f, g \in C^{1,2}([0,\infty) \times \mathbb{R}^d; \mathbb{R})$ the process

$$M_t^f := f(t, X_t) - f(0, X_0) - \int_0^t (\frac{\partial}{\partial s} + \mathcal{A}'_s) f(s, X_s) ds$$

is a continuous local martingale and

$$\langle M^f, M^g \rangle_t = \sum_{i=1}^d \sum_{k=1}^d \int_0^t a_{ik}(s, X_s) \frac{\partial}{\partial x_i} f(s, X_s) \frac{\partial}{\partial x_k} g(s, X_s) ds.$$

Finally, if the first derivative of f is bounded, then for every $0 \leq T < \infty$:

$$\forall y \in C([0,\infty)^d), t \in [0,T]: \quad \|\sigma(t,y)\| \overset{???}{\leq} K_T.$$

So M^f is continuous square integrable martingale.

Remark 5.2

If we set $b_i \equiv 0$ and $\sigma_{ij} \equiv \delta_{ij}$ for every $i, j = 1, ..., d$ then

$$\mathcal{A} f = \frac{1}{2} f = \frac{1}{2} \sum_{i=1}^d \frac{\partial}{\partial x_i^2} f, \quad f \in C^2(\mathbb{R}^d, \mathbb{R}).$$

Remark 5.3

Problem:
A continuous and adapted process W is a d-dimensional Brownian motion if and only if

$$M_t^f = f(W_t) - f(W_0) - \frac{1}{2} \int_0^t \Delta f(W_s) ds$$

is a continuous local martingale for every $f \in C^2(\mathbb{R}^d)$.

5.2 Weak Solutions and the Martingale Problem

The problem above gives a characterization of a Brownian motion. The basic idea of the martingale problem is to employ M^f as before. For now, we consider b_i and σ_{ij} to be progressively measurable functionals. Let $(\mathcal{A}'_t)_{t \geq 0}$ be defined by

$$(\mathcal{A}'_t u)(y) := \frac{1}{2} \sum_{i=1}^d \sum_{k=1}^d a_{ik}(t,y) \frac{\partial^2}{\partial x_i \partial x_k} u(y(t)) + \sum_{i=1}^d b_i(t,y) \frac{\partial}{\partial x_i} u(y(t)), t \in [0,\infty), y \in C([0,\infty)^d), u \in C^2(\mathbb{R}^d).$$

Definition 5.4

A probability measure $\mathbb{P}$ on $(C([0,\infty)), \mathcal{B}(C([0,\infty))))$ is called solution to the martingale problem associated with $(\mathcal{A}'_t)_{t \geq 0}$, if under this probability measure

$$M_t^f = f(y(t)) - f(y(0)) - \int_0^t (\mathcal{A}'_s f)(y) ds, \quad \forall t \in [0,\infty),$$

is a $\mathcal{F}_t$ continuous and local martingale for every $f \in C^2(\mathbb{R}^d)$. Here, $\mathcal{F}_t := \mathcal{G}_{t+} = \bigcap_{s>t} \mathcal{G}_t$, although $(\mathcal{G}_t)_{t\geq 0}$ is the augmentation under $\mathbb{P}$ of the canonical filtration of

$$\mathcal{B}_t := \mathcal{B}_t(C([0,\infty)^d)) := \sigma(z(s) : s \in [0,t]) = \varphi_t^{-1}(\mathcal{B}(C([0,\infty)^d))),$$

where

$$\varphi_t : C([0,\infty)^d) \to C([0,\infty)^d), z \mapsto (\varphi_t z)(s) := z(t \wedge s).$$

We show that weak solutions to the functional SDE is equivalent to the local martingale problem. One direction has already been shown: If there is a weak solution $((X,W),(\Omega,\mathcal{F},\mathbb{P}),(\mathcal{F}_t)_{t\geq 0})$ then with Proposition 5.1

$$M_t^f := f(t,X_t) - f(0,X_0) - \int_0^t \left(\frac{\partial}{\partial s} + \mathcal{A}'_s\right) f(s,X_s)ds, \quad \forall f \in C^{1,2}([0,\infty)\times\mathbb{R}^d)$$

is a continuous local martingale, i.e. a solution to the local martingale problem. The converse holds true with the next proposition.

Proposition 5.5

Let $\mathbb{P}$ be a probability measure on $(C([0,\infty)^d), \mathcal{B}(C([0,\infty)^d))$ under which the process M^f is a continuous local $\mathcal{F}_t$-martingale where $f(x) = x_i$ or $f(x) = x_i x_k$ for $i,k = 1,...,d$. Then there exists a Brownian motion W wrt. $(\tilde{\mathcal{F}}_t)_{t\geq 0}$ defined on the extension $(\tilde{\Omega},\tilde{\mathcal{F}},\tilde{\mathbb{P}})$ of $(C([0,\infty)), \mathcal{B}(C([0,\infty)^d)),\mathbb{P})$ such that $((X_t := y(t), W_t),(\tilde{\Omega},\tilde{\mathcal{F}},\tilde{\mathbb{P}}),(\tilde{\mathcal{F}}_t))$ is a weak solution to the functional SDE.

Proof. For $f(x) = X_i$ the differential operator is $(\mathcal{A}'f)(x) = b_i(s,X_s)$. Then by assumption

$$M_t^i := X_t^i - X_0^i - \int_0^t b_i(s,X_s)ds,$$

is a continuous local $\mathcal{F}_t$-martingale under $\mathbb{P}$. We now want to show that

$$\forall t \geq 0 : \quad \langle M^i, M^k\rangle_t = \int_0^t a_{ik}(s,X_s)ds \quad \mathbb{P}-a.s.$$

We do this by showing that $(M_t^i M_t^k - \int_0^t a_{ik}(s, X_s)ds)_{t\geq 0}$ is a continuous local martingale. See that

$$\begin{aligned}
&\int_0^t (X_s^i - X_t^i)b_k(s, X_s)ds + \int_0^t X_s^k - X_t^k)b_i(s, X_s)ds + \int_0^t b_i(s, X_s)ds \int_0^t b_i(s, X_s)ds \\
&= \int_0^t (M_s^i + X_0^i + \int_0^s b_i(l, X_l)dl - (M_t^i + X_0^i + \int_0^t b_i(l, X_l)dl))b_k(s, X_s)ds \\
&+ \int_0^t (M_s^k - X_0^k + \int_0^s b_k(l, X_l)dl - (M_t^k + X_0^k + \int_0^t b_k(l, X_l)dl))b_i(s, X_s)ds \\
&+ \int_0^t b_i(s, X_s)ds \int_0^t b_k(s, X_s)ds \\
&= \int_0^t (M_s^i - M_t^i)b_k(s, X_s)ds + \int_0^t (M_s^k - M_t^k)b_i(s, X_s)ds \\
&\overset{(*)}{=} M_t^i B_t^k - \int_0^t B_s^i dM_s^k - M_t^i \int_0^t dB_s^k + M_t^k B_t^i - \int_0^t B_s^k dM_s^i - M_t^k \int_0^t dB_s^i \\
&= M_t^i B_t^k + M_t^k B_t^i - M_t^i \int_0^t dB_s^k - M_t^k \int_0^t dB_s^i - \int_0^t B_s^i dM_s^k - \int_0^t B_s^k dM_s^i \\
&= M_t^i B_t^k + M_t^k B_t^i - M_t^i B_t^k - M_t^k B_t^i - \int_0^t B_s^i dM_s^k - \int_0^t B_s^k dM_s^i \\
&= -\left(\int_0^t \int_0^s b(l, X_l)dl dM_s^k + \int_0^t \int_0^s b(l, X_l)dl dM_s^i \right). \\
&(HierSindINdexfehler???)
\end{aligned} \tag{5.1}$$

although we used in $(*)$ the product formula for $dB_t^i = \int_0^t b(s, X_s)ds$:

$$M_t^i B_t^i = \int_0^t M_s^i dB_s^i + \int_0^t B_s^i dM_s^i + \underbrace{\langle M_t^i, B_t^i \rangle}_{=0}.$$

The last term is zero, as $(B_t^i)_{t\geq 0}$ is of finite variation. Because of (???) it holds that

$$\forall i = 1, ..., d: \quad \mathbb{P}(\int_0^t |b_i(s, X_s)|ds < \infty, t \geq 0) = 1,$$

which implies that (5.1) is a local martingale. One can show that

$$\begin{aligned}
M_t^i M_t^k - \int_0^t a_{ik}(s, X_s)ds &= M_t^{i,k} - X_0^i M_t^k - X_0^k M_t^i \\
&+ \int_0^t X_s^i dB_s^k - X_t^i B_t^k + \int_0^t X_s^k dB_s^i - X_t^k B_t^i + B_t^i B_t^k \\
&= M_t^{i,k} - X_0^i M_t^k - X_0^k M_t^i \\
&+ \int_0^t (X_s^i - X_t^i)b_i(s, X_s)ds + \int_0^t (X_s^k - X_t^k)b_k(s, X_s)ds + \int_0^t b_i(s, X_s)ds \int_0^t b_k(s, X_s)ds
\end{aligned}$$

This yields

$$\forall t \geq 0: \quad \langle M^i, M^k \rangle_t = \int_0^t a_{ik}(s, X_s)ds \quad \mathbb{P} - a.s.$$

Because the cross variation $\langle M^i, M^j \rangle_t$ is $\mathbb{P}$-a.s. an absolutely continuous function there exists with Theorem 2.5 and extension $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbb{P}})$ of $(C([0, \infty)^d), \mathcal{B}(C([0, \infty)^d)), \mathbb{P})$ endowed with the filtration $(\mathcal{F}_t)_{t\geq 0}$ which

satisfies the usual conditions and a matrix valued process $\rho := (\rho_{ij}(t))_{t\geq 0}$ and a Brownian motion $\tilde{W}$ such that

$$\forall t \geq 0, i, j = 1, ..., d : \tilde{P}(\int_0^t \rho_{ij}^2(s)ds < \infty) = 1$$

and

$$\forall t \geq 0, i = 1, ..., d : M_t^i = \sum_{j=1}^d \int_0^t \rho_{ij}(s)d\tilde{W}_s^j \quad \tilde{\mathbb{P}} - a.s.$$

Rewriting the definition of M_t^i yields

$$X_t^i = M_t^i + X_0^i + \int_0^t b_i(s, X_s)ds = X_0^i + \int_0^t b_i(s, X_s)ds + \sum_{j=1}^d \int_0^t \rho_{ij}(s)d\tilde{W}_s^j, \quad t \geq 0.$$

We now want to construct another Brownian motion on $(\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t)_{t\geq 0}, \tilde{\mathbb{P}})$ such that

$$\int_0^t \rho_{ij}(s)d\tilde{W}_s^i = \int_0^t \sigma(s, X_s)dW_s^i, \quad t \geq 0, i, j = 1, ..., d \quad \tilde{\mathbb{P}} - a.s.$$

There exists

$$R \in D := \{(\rho, \sigma) \mid \rho \text{ and } \sigma \text{ are } (d \times d) \text{ matrices: } \quad \rho\rho^T = \sigma\sigma^T\},$$

such that

$$\sigma = \rho R(\rho, \sigma) \quad \text{and} \quad R(\rho, \sigma)R^T(\rho, \sigma) = Id.$$

So R is an orthogonal matrix. Define

$$W_t := \int_0^t R^T(\rho(s), \sigma(s, X_s))d\tilde{W}_s, \quad t \geq 0$$

Because W^i is a continuous local martingale and

$$\begin{aligned}
\langle W^i, W^j\rangle_t &= \sum_{k=1}^d \sum_{l=1}^d \langle \int_0^\cdot R^T(\rho(s), \sigma(s, X_s))_{ik}d\tilde{W}_s^k, \int_0^\cdot R^T(\rho(s), \sigma(s, X_s))_{jl}d\tilde{W}_s^l\rangle_t \\
&\overset{???}{=} \sum_{k=1}^d \sum_{l=1}^d \int_0^t R^T(\rho(s), \sigma(s, X_s))_{ik}R^T(\rho(s), \sigma(s, X_s))_{jl}d\langle\tilde{W}^i, \tilde{W}^l\rangle_s \\
&= \sum_{k=1}^d \sum_{l=1}^d \int_0^t R^T(\rho(s), \sigma(s, X_s))_{ik}R^T(\rho(s), \sigma(s, X_s))_{jl}\delta_{jl}ds \\
&= t
\end{aligned}$$

it is a Brownian motion. Because

$$\forall t \geq 0, y \in C([0, \infty)^d) : \quad a_{ik}(t, y) := \sum_{j=1}^r \sigma_{ij}(t, y)\sigma_{kj}(t, y),$$

it follows that

$$\begin{aligned}
\int_0^t a_{ik}(s, X_s)ds = \langle M^i, M^k\rangle_t &= \sum_{j=1}^d \sum_{l=1}^d \int_0^t \rho_{ij}(s)\rho_{kl}(s)d\langle\tilde{W}^i, \tilde{W}^j\rangle_s \\
&= \sum_{j=1}^d \sum_{l=1}^d \int_0^t \rho_{ij}(s)\rho_{kl}(s)\delta_{jl}ds \\
&= \sum_{j=1}^d \int_0^t \rho_{ij}(s)\rho_{kj}(s)ds \quad \tilde{\mathbb{P}} - a.s.
\end{aligned}$$

It follows that

$$\forall i,k=1,...,d: \quad \tilde{\mathbb{P}}(\sum_{j=1}^{d}\rho_{ij}(t)\rho_{kj}(t)=a_{ik}(t,X_s),a.e.t\geq 0)=1,$$

which finally yields with (5.2) that

$$\forall t\geq 0, i=1,...,d, j=1,...,r: \quad \tilde{\mathbb{P}}(\int_0^t \sigma_{ij}^2(s,X_s)ds<\infty)=1.$$

Thus,

$$\forall t\geq 0, i=1,...,d, j=1,...,r: \quad \tilde{\mathbb{P}}(\int_0^t |b_i(s,X_s)|+\sigma_{ij}^2(s,X_s)ds<\infty)=1$$

and

$$\forall t\geq 0, i=1,...,d: \quad X_t^i=X_0^i+\int_0^t b_i(x,X_s)ds+\sum_{j=1}^{r}\int_0^t \sigma_{ij}(s,X_s)dW_s^j.$$

So $((\tilde{\Omega},\tilde{\mathcal{F}},\tilde{\mathbb{P}}),(X,W),(\mathcal{F}_t)_{t\geq 0})$ is a weak solution. □

Corollary 5.6

The existence of a solution $\mathbb{P}$ to the local martingale problem associated with $(\mathcal{A}_t')_{t\geq 0}$ is equivalent to the existence of a weak solution $((\tilde{\Omega},\tilde{\mathcal{F}},\tilde{\mathbb{P}}),(X,W),(\mathcal{F}_t)_{t\geq 0})$ to the functional SDE.
Both solutions are related by

$$\mathbb{P}=\tilde{\mathbb{P}}\circ X^{-1},$$

i.e. X induces a probability measure $\mathbb{P}$ on $(C([0,\infty)^d)),\mathcal{B}(C([0,\infty)^d)))$.

Corollary 5.7

The solution $\mathbb{P}$ to the local martingale problem with fixed but arbitrary initial distribution

$$\forall A\in\mathcal{B}(\mathbb{R}^d): \quad \mathbb{P}(X\in C([0,\infty)^d),X\in A)=\mu(A),$$

being unique is equivalent to the uniqueness in sense of law for the SDE

$$dX_t=b(t,X)dt+\sigma(t,X)dW_t, \quad t\geq 0.$$

Given progressively measurable functionals $b_i,\sigma_{ij}:[0,\infty)\times C([0,\infty)^d)\to\mathbb{R}$, for $i=1,...,d, j=1,...,r$, $(\mathcal{A}_t')_{t\geq 0}$ the associated family of differential operators and μ a probability measure on $\mathcal{B}(\mathbb{R}^d)$ consider the conditions

A) There exists a weak solution to the functional SDE with initial distribution μ.

B) There exists $\mathbb{P}$ solution to the local martingale problem associated with $(\mathcal{A}_t')_{t\geq 0}$ with

$$\forall A\in\mathcal{B}(\mathbb{R}^d): \quad \mathbb{P}(X_0\in A)=\mu(A).$$

C) There exists $\mathbb{P}$ a solution to the martingale problem associated with $(\mathcal{A}_t')_{t\geq 0}$ with

$$\forall A\in\mathcal{B}(\mathbb{R}^d): \quad \mathbb{P}(X_0\in A)=\mu(A).$$

Definition 5.8

A probability measure $\mathbb{P}$ on $(C([0,\infty)^d), \mathcal{B}(C([0,\infty)^d)))$ under which

$$M_t^f := f(y(t)) - f(y(0)) - \int_0^t (\mathcal{A}'_s f)(y) ds,$$

is a continuous $\mathcal{F}_t$-martingale, for every $f \in C_0^2(\mathbb{R}^d)$ is called solution to the martingale problem associated with $(\mathcal{A}'_t)_{t\geq 0}$.

Proposition 5.9

A) and B) are equivalent. C) implies A and B. Further, if either

- For every $T \in (0,\infty)$

$$\exists K_T, \forall t \in [0,T], y \in C([0,\infty)^d): \quad \|\sigma(t,y)\| \leq K_T$$

- For every σ_{ij} exists $\tilde{\sigma}_{ij} : [0,\infty) \times \mathbb{R}^d \to \mathbb{R}$ Borel measurable and bounded on compact sets such that

$$\forall t \geq 0, y \in C([0,\infty)^d): \quad \sigma_{ij}(t,y) = \tilde{\sigma}_{ij}(t, y(t)),$$

then it holds that A) implies C).

Proof. The equivalence of A) and B) has been already established. Let We show that C) implies A). Let C) be true. In the local martignale problem $f \in C^2(\mathbb{R}^d)$. So if $\mathbb{P}$ is solution to the martignale problem, it does not need to have compact support. Define the localizing sequence

$$S_k := \inf\{t \geq 0 \mid \|y(t)\| \geq k\},$$

i.e. $S_k \to \infty$, as $k \to \infty$. Because the canonical path process $X_t := y(t)$ posseses continuous paths, for every $t \leq S_k$:

$$y(t) \in \{x \in \mathbb{R}^d \mid \|x\| \leq k\} =: A.$$

Choose $g_k \in C_0^2(\mathbb{R}^d)$ such that for every $x \in A$ holds that $g_k(x) = f(x)$. Then, for every $t \geq 0$:

$$\begin{aligned} M^f_{t\wedge S_k} &= f(y(t\wedge S_k)) - f(y(0)) - \int_0^{t\wedge S_k} (\mathcal{A}f)(s,y(s)) ds \\ &= g_k(y(t\wedge S_k)) - g_k(y(0)) - \int_0^{t\wedge S_k} (\mathcal{A}g_k)(s,y(s)) ds = M^{g_k}_{t\wedge S_k}. \end{aligned}$$

As C) is ture, M^{g_k} is a martignale for every $k \geq 0$. Doobs optimal stopping implies that $(M^{g_k}_{t\wedge S_k})_{t\geq 0}$ is a martignale. So $(M^f_{t\wedge S_k})_{t\geq 0}$ is a martingale and thus M^f a continuous local martingale. □

SECTION 6

STRONG MARKOV PROPERTY FOR THE SOLUTION OF SDES

We consider the time-homogenous formulation of the SDE

$$X_t = x + \int_0^t b(X_s)ds + \int_0^t \sigma(X_s)dW_s, \quad t \in [0,\infty),$$

where $b_i, \sigma_{ij} : \mathbb{R}^d \to \mathbb{R}$ a time-homogenous and borel measurable.

Remark 6.1

This still allows for time dependence, because we can append the time to the state variable as follows: $X_t^{d+1} = t$. Then the equation is time homogenous if we choose $b_{d+1}(x) = 1$ and $\sigma_{d+1}(x) = 0$. This augmentation makes the diffusion not of full rank, i.e. degenerate. Some results require the non-degeneracy.

Definition 6.1

The SDE is well posed, if for every $x \in \mathbb{R}^d$, there exists a weak solution, that is unique in sense of its probability law.

We know that there is a strong solution if b and σ are Lipschitz and satisdy the linear growth condition. If σ is the identity matrix and b is uniformly bounded, then it is also well posed. Later we shall find less restrictive conditions.

Remark 6.2

If the SDE is well posed, then X is a weak solution with inital condition $X_0 = x$ and it induces a measure $\mathbb{P}^X$ on $(C([0,\infty)^d), \mathcal{B}(C([0,\infty)^d)))$ is the coordinate mapping process ($\omega(t)$???) on this measurable space, the filtration $(\mathcal{B}_t)_{t\geq 0}$ and $(\mathbb{P}^x)_{x\in\mathbb{R}^d}$ a strong Markov family. The time homogenous operator is

$$(\mathcal{A}f)(x) = \frac{1}{2}\sum_{i,k=1}^{d} a_{ik}(x)\frac{\partial^2}{\partial x_i \partial x_j}f(x) + \sum_{i=1}^{d} b_i(x)\frac{\partial}{\partial x_i}f(x).$$

Definition 6.2

Assume $b_i, \sigma_{ij} : \mathbb{R}^d \to \mathbb{R}$, $i = 1,...,d, j = 1,...,r$ are bounded on compact subsets of $\mathbb{R}^d$. We call a proability measure $\mathbb{P}$ on $(C([0,\infty)^d), \mathcal{B}(C([0,\infty)^d)))$ under which for all $s,t \in [0,\infty), s < t, f \in C_0^2(\mathbb{R}^d)$:

$$\mathbb{E}[f(y(t)) - f(y(0)) - \int_0^t (\mathcal{A}f)(y(u))du \mid \mathcal{B}_s] = 0, \quad \mathbb{P}-a.s.$$

a solution of the time-homogenous martingale problem. We denote by $\mathbb{P}^x$ any solution to this problem if it satisfies

$$\mathbb{P}^x(y \in C([0,\infty)^d), y(0) = x) = 1.$$

The time-homogenous martingale problem is well posed if for every $x \in \mathbb{R}^d$, there is exactly one solution $\mathbb{P}^x$.

Remark 6.3

Note that in the definition above we replaced $\mathcal{F}_t$ with $\mathcal{B}_t$. This is allowed due to the following: Assume either

- $$\forall T \in (0,\infty) \exists t \in [0,T], y \in C([0,\infty)^d): \quad \|b_i(t,y)\| + \|\sigma_{ij}(t,y)\| \leq K_T$$
- There exists $\tilde{b}_i, \tilde{\sigma}_{ij} : [0,\infty) \times \mathbb{R}^d \to \mathbb{R}$ bounded on compact sets such that $b_i(t,y) = \tilde{b}_i(t,y(t))$ and $\sigma_{ij}(t,y) = \tilde{\sigma}_{ij}(t,y(t))$

A probability measure on $(C([0,\infty)^d), \mathcal{B}(C([0,\infty)^d)))$, with filtration $(\mathcal{F}_t)_{t\geq 0}$, where $\mathcal{F}_t := \bigcap_{s>t} \mathcal{F}_t$ and $\mathcal{B}_t := \mathcal{B}_t(C([0,\infty)^d))$, $f \in C_0^2(\mathbb{R}^d)$ satisfies that M_t^f a $\mathcal{B}_t$-martingale is also a $\mathcal{F}_t$-martingale.

Remark 6.4

The well posedness of the time homogenous martingale problem is equivalent to the well posedness of the time homogenous integral equation. This follows from the previous section and setting $X_t^{d+1} = t$.

Lemma 6.3

For every bounded stopping time of the filtration $(B_t)_{t\geq 0}$ it holds that

$$B_T = \sigma(y(t \wedge T) : t \in [0,\infty)).$$

Proof. " $\subseteq$ "
A random variable Z is $\mathcal{B}_T$ measurable if and only if

$$\forall s \geq 0, A \in ??? : \quad \{Z \in A\} \cap \{T \leq s\} \in \mathcal{B}_s.$$

We have $z := y(t \wedge T)$ on $(C([0,\infty)^d), \mathcal{B}(C([0,\infty)^d)))$.
Case 1: $t \leq s$: Because $t \wedge T \leq t$ and its $\mathcal{B}_t$ measurable it follows that $y(t \wedge T)$ is $\mathcal{B}_t$ measurable. THen

$$\underbrace{\{y(t \wedge T) \in A\}}_{\in \mathcal{B}_t \subseteq \mathcal{B}_s} \cap \{T \leq s\} \in \mathcal{B}_s.$$

Case 2: $t > s$: We have $t \wedge T = T = s \wedge T$. So

$$\{y(t \wedge T) \in A\} \cap \{T \leq s\} = \{y(s \wedge T) \in A\} \cap \{T \leq s\} \in \mathcal{B}_s.$$

" $\supseteq$ "
We need to show that $\mathcal{B}_T \subseteq \sigma(y(t \wedge T); t \in [0,\infty))$. For every $t \geq 0$ let

$$\varphi_t : C([0,\infty)^d) \to C([0,\infty)^d), y \mapsto \varphi_t y, \quad (\varphi_t y)(s) = y(t \wedge s), \quad s \in [0,\infty).$$

Because by construction for every $s \in [0, T(y)]$ it holds that

$$y(s) = (\varphi_{T(y)} y)(s)$$

it follows that for every $y \in C([0,\infty)^d)$ it holds that

$$T(y) = T(\varphi_{T(y)}(y)).$$

By setting $t := T(y)$ and for $A \in \mathcal{B}_T$ we have that

$$y \in A \iff y \in (A \cap \{T \leq t\}) \overset{(*)}{\iff} \varphi_t y \in (A \cap \{T \leq t\}) \iff \varphi_t y \in A,$$

although $(*)$ follows form the fact that $A \cap \{T \leq t\} \in \mathcal{B}_t$ and again that

$$s \in [0,t]: \quad y(s) = (\varphi_t(y))(s),$$

which we already used earlier. Finally, we have

$$A = \{y \in C([0,\infty)^d) \mid \varphi_{T(y)}(y) \in A\} = \{y \in C([0,\infty)^d) \mid y(\cdot \wedge T(y)) \in A\}$$

□

Lemma 6.4

Let T be a bounded stopping time of $(\mathcal{B}_t)_{t\geq 0}$ and $\mathcal{G}$ a countable determined sub σ-algebra of $\mathcal{B}_T$ such that $y(T)$ is $\mathcal{G}$-measurable. Suppose that b and σ are bounded on compact subsets of $\mathbb{R}^d$ and a probability measure $\mathbb{P}$ on $(\Omega, \mathcal{B}) := (C([0,\infty)^d), \mathcal{B}(C([0,\infty)^d)))$ solves the time-homogenous martingale problem. Denote by

$$\mathbb{P}[\cdot \mid \mathcal{G}](\cdot) : \Omega \times \mathcal{B} \to [0,1]$$

the regular conditional probability given $\mathcal{G}$. Then there exists a $\mathbb{P}$-null set $N \in \mathcal{G}$ such that

$$\forall \omega \notin N: \quad \mathbb{P}_\omega := \mathbb{P}[\cdot \mid \mathcal{G}](\omega) \circ \theta_T^{-1}$$

solves the time-homogenous martingale problem with $x = \omega(T(\omega))$, although

$$\theta_T : C([0,\infty)^d) \to C([0,\infty)^d), \omega \mapsto \theta_T(\omega),$$

is a measurable mapping such that for all $t \geq 0$ holds that $(\theta_T(\omega))(t) = \omega(T(\omega) + t)$.

Proof. ...???

□

Theorem 6.5: Strong Markov Property

Suppose the coefficients b and σ are bounded on compact subsets of $\mathbb{R}^d$ and the time-homogenous martingale problem is well posed. Then for every $\mathcal{B}_t$-stopping time T, $F \in \mathcal{B}(C([0,\infty)^d))$ and $x \in \mathbb{R}^d$:

$$\mathbb{P}^x[\theta_T^{-1}(F) \mid \mathcal{B}_T](\omega) = \mathbb{P}^{\omega(T)}(F), \quad \mathbb{P}^x - a.s. \text{ on } \{T < \infty\}.$$

Theorem 6.6

Let $b_i, \sigma_{ij} : \mathbb{R}^d \to \mathbb{R}$ be bounded and continuous and the time-homogenous SDE

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t.$$

For every initial distribution μ on $\mathcal{B}(\mathbb{R}^d)$ such that

$$\exists m > 1: \quad \int_{\mathbb{R}^d} \|x\|^{2m} d\mu(x) < \infty,$$

exists a weak solution.

Definition 6.7

A collection $\mathcal{D} \ni \varphi : \mathbb{R}^d \to \mathbb{R}$ of Borel measurable functions is called determining class on $\mathbb{R}^d$ if for any finite measures μ_1 and μ_2 on $\mathcal{B}(\mathbb{R}^d)$

$$\forall \varphi \in \mathcal{D}: \quad \int_{\mathbb{R}^d} \varphi(x) d\mu_1(x) = \int_{\mathbb{R}^d} \varphi(x) d\mu_2(x) \quad \Longrightarrow \quad \mu_1 = \mu_2.$$

Remark 6.5

The collection $C_0^\infty(\mathbb{R}^d)$ is a determining class on $\mathbb{R}^d$.

Lemma 6.8

Suppose for every $f \in C_0^\infty(\mathbb{R}^d)$ the Cauchy problem

$$\frac{\partial}{\partial t} u(t,x) = (\mathcal{A}u)(t,x), (t,x) \in (0,\infty) \times \mathbb{R}^d, \quad u(0,x) = f(x), x \in \mathbb{R}^d$$

has a solution $u_f \in C([0,\infty)^d \times \mathbb{R}^d) \cap C^{1,2}((0,\infty) \times \mathbb{R}^d)$ that is bounded on $[0,T] \times \mathbb{R}^d$ for every $T > 0$.
If $\mathbb{P}^x$ and $\tilde{\mathbb{P}}^x$ are solutions of the time homogenous martignale problem with initial condition $x \in \mathbb{R}^d$, their one dimensional distributions agree, i.e.

$$\forall t \in [0,\infty), A \in \mathcal{B}(\mathbb{R}^d): \quad \mathbb{P}^x(y(t) \in A) = \tilde{\mathbb{P}}^x(y(t) \in A).$$

We want to go beyond the uniqueness of the one dimensional marginals to that of all finite-dimensional distributions.

Remark 6.6

We show this via induction.
$n = 1$: $\mathbb{P}^x \circ y(t_1)^{-1} = \tilde{\mathbb{P}}^x \circ y(t_1)^{-1}$.
IS: Consider $0 \leq t_1 < ... < t_{n+1}$ and $f_i \in C_0^2(\mathbb{R}^d)$. With the tower property we have

$$\begin{aligned}
\mathbb{E}^{\mathbb{P}^x}[\prod_{i=1}^{n+1} f_i(y(t_i))] &= \mathbb{E}^{\mathbb{P}^x}[\mathbb{E}^{\mathbb{P}^x}[\prod_{i=1}^{n+1} f_i(y(t_i)) \mid \mathcal{B}_{t_n}]] \\
&= \mathbb{E}^{\mathbb{P}^x}[\prod_{i=1}^{n} f_i(y(t_i)) \underbrace{\mathbb{E}^{\mathbb{P}^x}[f_{n+1}(y(t_{n+1})) \mid \mathcal{B}_{t_n}]}_{=:u(t_{n+1})}]
\end{aligned}$$

Because

$$y(t_{n+1})(\omega) = \omega(t_n + (t_{n+1} - t_n)) = (\theta_{t_n}(\omega))(t_{n+1} - t_n) = y(t_{n+1} - t_n)(\theta_{t_n}(\omega))$$

it follows that

$$\begin{aligned}
u(t_{n+1})(\omega) &= \int_\Omega f_{n+1}(y(t_{n+1})(\omega'))d\mathbb{P}^x[\omega' \mid \mathcal{B}_{t_n}](\omega) \\
&= \int_\Omega f_{n+1}(y(t_{n+1}-t_n)(\theta_{t_n}(\omega')))d\mathbb{P}^x[\omega' \mid \mathcal{B}_{t_n}](\omega) \\
&= \int_\Omega f_{n+1}(y(t_{n+1}-t_n)(\omega'))d\left(\mathbb{P}^x[\cdot \mid \mathcal{B}_{t_n}](\omega)\circ\theta_{t_n}^{-1}\right)(\omega') \\
&= \int_\Omega f_{n+1}(y(t_{n+1}-t_n)(\omega'))d\mathbb{P}^{y(t_n)(\omega)}(\omega') \\
&= \mathbb{E}^{\mathbb{P}^{y(t_n)(\omega)}}[f_{n+1}(y(t_{n+1}-t_n))].
\end{aligned}$$

Thus,

$$\begin{aligned}
\mathbb{E}^{\mathbb{P}^x}[\prod_{i=1}^{n+1} f_i(y(t_i))] &= \mathbb{E}^{\mathbb{P}^x}[\prod_{i=1}^{n} f_i(y(t_i))\mathbb{E}^{\mathbb{P}^{y(t_n)(\cdot)}}[f_{n+1}(y(t_{n+1}-t_n))]] \\
&\overset{IV}{=} \mathbb{E}^{\tilde{\mathbb{P}}^x}[\prod_{i=1}^{n} f_i(y(t_i))\mathbb{E}^{\mathbb{P}^{y(t_n)(\cdot)}}[f_{n+1}(y(t_{n+1}-t_n))]] \\
&\overset{1D-marginals}{=} \mathbb{E}^{\tilde{\mathbb{P}}^x}[\prod_{i=1}^{n} f_i(y(t_i))\mathbb{E}^{\tilde{\mathbb{P}}^{y(t_n)(\cdot)}}[f_{n+1}(y(t_{n+1}-t_n))]] \\
&= \mathbb{E}^{\tilde{\mathbb{P}}^x}[\prod_{i=1}^{n+1} f_i(y(t_i))]
\end{aligned}$$

Because $C_0^2((\mathbb{R}^d)^n)$ are a determining class on $(\mathbb{R}^d)^n$ and we have shown that

$$\int_{(\mathbb{R}^d)^n}\prod_{i=1}^n f_i(z_i)d(\mathbb{P}^x\circ(y(t_1),...,y(t_n))^{-1})(z_1,...,z_n) = \int_{(\mathbb{R}^d)^n}\prod_{i=1}^n f_i(z_i)d(\tilde{\mathbb{P}}^x\circ(y(t_1),...,y(t_n))^{-1})(z_1,...,z_n)$$

it follows that $\mathbb{P}^x$ and $\tilde{\mathbb{P}}^x$ have the same finite dimensional distribution. We know that $\mathbb{P}^x \circ (y(t_1),...,y(t_n))^{-1}$ uniquely determine $\mathbb{P}^x$ on $\mathcal{B}(C([0,\infty)^d))$, which is well known. So the infinte dimensional distributions $\mathbb{P}^x$ and $\tilde{\mathbb{P}}^x$ are equal.

Proposition 6.9

Suppose for every $x \in \mathbb{R}^d$ any two solutions $\mathbb{P}^x$ and $\tilde{\mathbb{P}}^x$ of the time homogenous martingale problem with initial condition x have the same one dimensional marginal distribution, i.e.

$$\forall t \in [0,\infty), A \in \mathcal{B}(\mathbb{R}^d): \quad \mathbb{P}^x(y(t) \in A) = \tilde{\mathbb{P}}^x(y(t) \in A).$$

Then, for every $x \in \mathbb{R}^d$ the solution to the time homogenous martignale problem with initial condition $x \in \mathbb{R}^d$ is unique.

Theorem 6.10

Suppose b_i and σ_{ij} are bounded on compact subsets of $\mathbb{R}^d$ and such that for every $f \in C_0^\infty(\mathbb{R}^d)$ the Cauchy problem has a solution $u_f \in C([0,\infty)\times\mathbb{R}^d) \cap C^{1,2}((0,\infty)\times\mathbb{R}^d)$ that is bounded on

$[0,T] \times \mathbb{R}^d$ for every $T > 0$. Then for every $x \in \mathbb{R}^d$ there is at most one solution tot the time homogenous martingale problem.

Corollary 6.11

Let b_i and σ_{ij} be bounded, continuous such that for every $f \in C_0^\infty(\mathbb{R}^d)$ the Cauchy problem has a solution $u_f \in C([0,\infty) \times \mathbb{R}^d) \cap C^{1,2}((0,\infty) \times \mathbb{R}^d)$ that is bounded on $[0,T] \times \mathbb{R}^d$ for every $T > 0$. Then the time homogenous martingale problem is well posed.

SECTION 7

MARKOV PROCESSES

Definition 7.1

A mapping $P : [0,\infty) \times E \times \mathcal{E} \to [0,1]$ is called time homogeneous (sub)-Markov transition function on $(E,\mathcal{E})$ if

- $P(t,\cdot,\cdot)$ is a (sub-) Markov kernel from $(E,\mathcal{E})$ to $(E,\mathcal{E})$ for every $t \geq 0$.
- $\forall s,t \geq 0$: $P(s+t,\cdot,\cdot) = P(t,\cdot,\cdot) \circ P(s,\cdot,\cdot)$.

P is called normal, if additionally $\forall x \in E$: $\{x\} \in \mathcal{E}$ and $P(0,x,\{x\}) = 1$.

Definition 7.2

Let $(\Omega,\mathcal{F},(\mathcal{F}_t)_{t\geq 0})$ be filtered space $(E,\mathcal{B}(E))$ polish space and $(\mathbb{P}_x)_{x\in E}$ a family of probability measures on $(\Omega,\mathcal{F})$. A $(\mathcal{F}_t)$-adapted E-valued process X is called Markov process if

- $\forall B \in \mathcal{F}_\infty^X$ the mapping $x \mapsto \mathbb{P}_x(B)$ is $\mathcal{B}(E)$-measurable.
- The weak Markov property holds: $\forall x \in E, A \in \mathcal{B}(E), s,t \geq 0$:

$$\mathbb{P}_x[X_{t+s} \in A \mid \mathcal{F}_s] = \mathbb{P}_{X_s}(X_t \in A) \quad \mathbb{P}_X - a.s.$$

X is called normal, if additionally $\{x\} \in \mathcal{B}(E)$ and $\mathbb{P}_X(X_0 = x) = 1$ $\forall x \in E$.

Theorem 7.3

Let X be a Markov process, then $P(t,x,A) := \mathbb{P}_X(X_t \in A)$, $x \in E, A \in \mathcal{B}(E), t \geq 0$ is a Markov transition function. Additionally,

$$X \text{ is normal} \quad \Longleftrightarrow \quad P \text{ is normal.}$$

Theorem 7.4

Let X be Levy process and set $P(t,x,A) := \mathbb{P}(X_t + x \in A)$ for every $t \geq 0, x \in \mathbb{R}^d, A \in \mathcal{B}(\mathbb{R}^d)$. Then for the respective sub-Markov semigroup $(P_t)_{t\geq 0}$ on $\mathcal{B}(\mathbb{R}^d)$ we have for every $t \geq 0$:

$$P_t C_0(\mathbb{R}^d) \subseteq C_0(\mathbb{R}^d).$$

Definition 7.5

A sub-Markov transition function P is called Feller transition function, if the respective semigroup $(P_t)_{t\geq 0}$ satisfies for every $t \geq 0$:

$$P_t C_b(E) \subseteq C_b(E).$$

Additionally, we call P a C_0 transition function if $\forall t \geq 0$:

$$P_t C_0(E) \subseteq C_0(E).$$

Theorem 7.6

Let E be locally compact, separable metric space and $(P_t)_{t\geq 0}$ a family of bounded, linear operators with $P_t : C_0(E) \to C_0(E)$ and

- $P_0 = I$ and $P_{s+t} = P_s \circ P_t \ \forall s,t \geq 0$
- $\forall t \geq 0$: $\|P_t\| \leq 1$
- $\forall f \in C_0(E)$ with $0 \leq f \leq 1$: $0 \leq P_t f \leq 1$

then $\exists\, P$ a C_0 transition function on $(E, \mathcal{B}(E))$ such that the respective semigroup on $C_0(E)$ coincides with $(P_t)_{t\geq 0}$.

Remark 7.1

$(X_t)_{t\geq 0}$ Markov process $\iff$ $P(t,x,A)$ $\iff$ $(P_t)_{t\geq 0}$.

Definition 7.7

A family $(P_t)_{t\geq 0}$ of bounded, linear operators on B is called semigroup linear contractions, if

- $P_0 = I$ and $\forall s,t \geq 0$: $P_{s+t} = P_s \circ P_t$
- $\forall t \geq 0$: $\|P_t\| \leq 1$.

Definition 7.8

Let $(P_t)_{t\geq 0}$ be a halfgroup linear contractions on B, then

$$B_0 := \{f \in B \mid \lim_{t \searrow 0} \|P_t f - f\| = 0\} \subseteq B$$

is called area of strong continuity. If $B_0 = B$ then we call $(P_t)_{t\geq 0}$ strong continuous contracting semigroup.

Definition 7.9

Let $(P_t)_{t\geq 0}$ be contracting semigroup on B. A linear operator $G : \mathcal{D}(G) \to B$ with

$$D(G) := \{f \in B \mid s - \lim_{t\searrow 0} \frac{1}{t}(P_t f - f) \text{ exists}\} \subseteq B$$

is called infinitesimal generator of $(P_t)_{t\geq 0}$ if

$$\forall f \in \mathcal{D}(G): \quad Gf = s - \lim_{t\searrow 0} \frac{1}{t}(P_t f - f).$$

Lemma 7.10

Let $(P_t)_{t\geq 0}$ be contracting semigroup on B then

- B_0 is Banach space
- $\forall t \geq 0$: $P_t B_0 \subseteq B_0$
- $f \in B_0 \iff \forall t \geq 0$: $P_{t+h} f \xrightarrow{s} P_t f$, $h \searrow 0$.

Proof. • B_0 is vector space, because for $f, g \in B$ and $a, b \in \mathbb{R}$:

$$\|P_t(af + bg) - (af + bg)\| \leq |a|\|P_t f - f\| + |b|\|P_t g - g\| \to 0, t \searrow 0$$

it follows $af + bg \in B_0$.
To show: B_0 is closed in B.
Let $(f_n)_{n\in\mathbb{N}} \subseteq B_0$ with $f_n \overset{strong}{\to} f \in B$, then

$$\begin{aligned}\|P_t f - f\| &= \|P_t f_n - f_n + P_t(f - f_n) - (f - f_n)\| \\ &\leq \|P_t f_n - f_n\| + \underbrace{\|P_t\|}_{\leq 1}\|f_n - f\| + \|f_n - f\| \\ &\leq \|P_t f_n - f_n\| + 2\|f_n - f\|.\end{aligned}$$

Because $f_n \in B_0$ it follows that

$$\limsup_{t\searrow 0} \|P - tf - f\| \leq 0 + 2\|f_n - f\| \to 0, n \to \infty$$

which implies $f \in B_0$.

- Let $f \in B_0$ and $t \geq 0$ then $\forall h \geq 0$:

$$\|P_h(P_t f) - P_t f\| = \|P_t(P_h f - f)\| \leq \|P_t\|\|P_h f - f\| \leq \|P_h f - f\|.$$

Thus,

$$\limsup_{h\searrow 0} \|P_h(P_t f) - P_t f\| = 0$$

and so $P_t f \in B_0$.

- " $\Rightarrow$ " follows with b)
" $\Leftarrow$ " set $t = 0$.

□

Remark 7.2

G is in general not a bounded operator

Definition 7.11

Let $(P_t)_{t\geq 0}$ be strong continuous contracting semigroup on B. Define

$$\forall 0 \leq a < b < \infty, f \in B: \quad B \ni \int_a^b P_s f ds := s - \lim_{n\to\infty} \sum_{k=1}^{n} (t_k - t_{k-1}) P_{s_k} f,$$

where $\forall n \in \mathbb{N}$, $a = t_0 < ... < t_n = b$ a partition of $[a,b]$ such that $\max_{k=1,...,n} |t_k - t_{k-1}| \to 0$, $n \to \infty$ and $s_k \in [t_{k-1}, t_k)$ for every $k = 1, ..., n$.

Lemma 7.12

Let $(P_t)_{t\geq 0}$ be strong continuous contracting semigroup on B, then

- $\forall f \in B, 0 \leq a < b < \infty$: $\| \int_a^b P_s f ds\| \leq \int_a^b \|P_s f\| ds$
- $\forall f \in B, t \geq 0$: $s - \lim_{h \searrow 0} \frac{1}{h} \int_t^{t+h} P_s f ds = P_t f$
- $\forall f \in B, 0 \leq a < b < \infty$: $P_t \left(\int_a^b P_s f ds \right) = \int_{a+t}^{b+t} P_s f ds.$

Proof.

- Because the norm $\|\cdot\|$ is continuous it follows

$$\| \int_a^b P_s f ds\| = \|s - \lim_{n\to\infty} \sum_{k=1}^{n} (t_k - t_{k-1}) P_{s_k} f\| \leq s - \lim_{n\to\infty} \sum_{k=1}^{n} (t_k - t_{k-1}) \|P_{s_k} f\| = \int_a^b \|P_s f\| ds.$$

- By strong continuity $\forall \epsilon > 0 \exists \delta > 0 \forall s, t \geq 0$ with $|s - t| < \delta$: $\|P_s f - P_t f\| < \epsilon$. Its application yields

$$\begin{aligned} \| \frac{1}{h} \int_t^{t+h} P_s f ds - P_t d\| &= \| \frac{1}{h} \int_t^{t+h} P_s f - P_t f ds\| \\ &\overset{a)}{\leq} \frac{1}{h} \int_t^{t+h} \|P_s f - P_t f\| ds < \epsilon. \end{aligned}$$

  As $\epsilon > 0$ arbitrary, the assertion follows.

- Because a bounded linear operator is continuous it follows:

$$\begin{aligned} P_t \left(s - \lim_{n\to\infty} \sum_{k=1}^{n} (t_k - t_{k-1}) P_{s_k} f \right) &= s - \lim_{n\to\infty} \sum_{k=1}^{n} (t_k - t_{k-1}) P_t \circ P_{s_k} f \\ &= s - \lim_{n\to\infty} \sum_{k=1}^{n} (t_k - t_{k-1}) P_{t+s_k} f \\ &= \int_a^b P_{s+t} f ds = \int_{a+t}^{b+t} P_s f ds. \end{aligned}$$

□

Theorem 7.13

Let G be the generator of a semigroup $(P_t)_{t\geq 0}$ linear contractions on B, then we have

- $\overline{\mathcal{D}(G)} = B_0$ and $G\mathcal{D}(G) \subseteq B_0$. Especially,

$$\forall f \in B_0, t \geq 0: \quad \int_0^t P_s f ds \in \mathcal{D}(G) \quad \text{and} \quad P_t f - f = G(\int_0^t P s f ds).$$

- $\forall f \in \mathcal{D}(G), t \geq 0$: $P_t f \in \mathcal{D}(G)$ and

$$\frac{d}{dt} P_t f := s - \lim_{h\to 0} \frac{1}{h}(P_{t+h} - Pf) = P_t G f = G P_t f.$$

  Especially,

$$P_t f - f = \int_0^t G P_s f ds = \int_0^t P_s G f ds.$$

- The operator G is closed, i.e. $\forall (f_n)_{n\in\mathbb{N}} \subseteq \mathcal{D}(G)$ with $f_n \xrightarrow{s} f \in B$ and $Gf_n \xrightarrow{s} g \in B$ it follows $f \in \mathcal{D}(G)$ and $Gf = g$.

Proof. • To show: $\mathcal{D}(G) \subseteq B_0$

Let $f \in \mathcal{D}(G)$, then

$$\exists c > 0, h_0 > 0 \forall h \leq h_0: \quad \|\frac{1}{h}(P_h f - f) - Gf\| < c.$$

Then $\forall \epsilon > 0$ and $t < \delta := \min\{h_0, \frac{\epsilon}{2}(c + \|f\|/h_0)\}$ yields

$$\begin{aligned}
\|P_t f - f\| &= t\|\frac{1}{t}(P_t f - f) - Gf - \frac{1}{h_0}(P_{h_0} f - f) + Gf + \frac{1}{h_0}(P_{h_0} f - f)\| \\
&\leq t(\|\frac{1}{t}(P_t f - f) - Gf\| + \|\frac{1}{h_0}(P_{h_0} f - f) - Gf\|) + \frac{t}{h_0}\|P_{h_0} f - f\| \\
&\leq 2tc + 2\frac{t}{h_0}\|f\| = 2t(c + \frac{\|f\|}{h_0}) < \epsilon.
\end{aligned}$$

Thus, $f \in B_0$.

To show: $\mathcal{D}(G)$ is dense in B_0

Let $f \in B_0$ then with the lemm above, it follows that

$$\frac{1}{h}(P_h - I)\int_0^t P_s f ds \overset{excercise}{=} \frac{1}{h}(\int_h^{t+h} P_s f ds - \int_0^t P_s f ds) = \frac{1}{h}\int_t^{t+h} P_s f ds - \frac{1}{h}\int_0^h P_s f ds.$$

Because the right hand side exists as a strong limit for $h \searrow 0$ it follows

$$\int_0^t P_s f ds \in \mathcal{D}(G) \quad \text{and} \quad G\int_0^t P_s f ds = s - \lim_{h\searrow 0}(P_h \int_0^t P_h f ds - \int_0^t P_s f ds) = P_t f - f.$$

Set $f_n := n\int_0^{1/n} P s f ds$, $n \in \mathbb{N}$, then $(f_n)_{n\in\mathbb{N}} \subseteq \mathcal{D}(G)$ (because $\mathcal{D}(G)$ is a linear space and $\int_0^t P_s f ds \in \mathcal{D}(G)$) and $s - \lim_{n\to\infty} f_n = f$, due to

$$\|f_n - f\| = \|n\int_0^{1/n} P_s f ds\| = \|n\int_0^{1/n} P_s f - f ds\| \leq n\int_0^{1/n} \|P_s f - f\| ds \to 0.$$

Thus, $\mathcal{D}(G)$ is tight in B_0.

To show: $G(\mathcal{D}(G)) \subseteq B_0$:

Let $f \in \mathcal{D}(G)$, then with $f = P_0 f \in B_0$. With the Lemma above we get $\frac{1}{t}(P_t f - f) \in B_0$ for all $t > 0$. Because B_0 is Banach space, that is closed it follows that

$$Gf = s - \lim_{t \searrow 0} \frac{1}{t}(P_t f - f) \in B_0.$$

- Let $f \in \mathcal{D}(G)$ and $t \geq 0$, then

$$\frac{1}{h}(P_h P_t f - P_t f) = \frac{1}{h}(P_t P_h f - P_t f) = P_t \frac{1}{h}(P_h f - f) \overset{s}{\to} P_t G f, \quad h \searrow 0,$$

although we used that P_t is bounded $\forall t \geq 0$ and thus the limit exists. So we have shown that $P_t f \in \mathcal{D}(G)$. Further, the above shows that $G P_t f = P_t G f$.
To show: $\forall t > 0$ and $f \in \mathcal{D}(G)$: $s - \lim_{h \searrow 0} \frac{1}{h}(P_t f - P_{t-h} f) = P_t G f$.
As $(P_t)_{t \geq 0}$ is a contracting semigroup:

$$\begin{aligned} \| \frac{1}{h}(P_t f - P_{t-h} f) - P_t G f \| &\leq \underbrace{\| P_{t-h} \|}_{\leq 1} \| \frac{1}{h}(P_h f - f) - P_h G f \| \\ &\leq \underbrace{\| \frac{1}{h}(P_h f - f) - G f \|}_{\to 0, \text{ by definition}} + \underbrace{\| P_h \underbrace{G f}_{\in B_0} - G f \|}_{\to 0, \text{ because } \overline{\mathcal{D}(G)} = B_0} \to 0. \end{aligned}$$

To show: $\forall t \geq 0$ and $f \in \mathcal{D}(G)$: $P_t f - f = \int_0^t P_s G f ds$.

$$\| P_t f - f - \int_0^t P_s G f ds \| \leq \underbrace{\| P_t f - f - \frac{1}{n}(P_h - I) \int_0^t P_s f ds \|}_{\overset{???}{\to} 0} + \underbrace{\| \int_0^t P_s (\frac{1}{h}(P_h f - f) - G f) ds \|}_{\leq \int_0^t \underbrace{\| P_s \|}_{\leq 1} \frac{1}{h} \| P_h f - f - G f \| ds} \to 0, h \searrow 0.$$

The above is equivalent to $P_t f - f = \int_0^t P_s G ds = \int_0^t G P_s f ds$.

- Let $(f_n)_{n \in \mathbb{N}} \subseteq \mathcal{D}(G)$ with $f_n \overset{s}{\to} f \in B$ and $G f_n \overset{s}{\to} g \in B$.
To show: $f \in \mathcal{D}(G)$ and $Gf = g$.

$$\begin{aligned} \| P_t f - f - \int_0^t P_s g ds \| &= \| P_t (f - f_n) - (f - f_n) + (P_t f_h - f_h) - \int_0^t P_s g ds \| \\ &\leq (\| P_t \| + 1) \| f_n - f \| + \| P_t f_h - f_h - \int_0^t P_s g ds \| \\ &\leq 2 \| f_n f \| + \| \int_0^t P_s G f_n - P_s g ds \| \\ &\leq 2 \underbrace{\| f_n - f \|}_{\overset{VSS}{\to} 0, n \to \infty} + \int_0^t \underbrace{\| P_s \|}_{\leq 1} \| G f_n - g \| ds \to 0, n \to \infty. \end{aligned}$$

Thus

$$s - \lim_{t \searrow 0} \frac{1}{t}(P_t f - f) \overset{above}{=} s - \lim_{t \searrow 0} \frac{1}{t} \int_0^t P_s g ds \overset{???}{=} g.$$

□

Definition 7.14

A family $(P_t)_{t\geq 0}$ of bounded linear operators on $\mathcal{B}(E)$ is called sub Markov semigroup if $\forall s,t\geq 0$:

- $P_t : \mathcal{B}(E) \to \mathcal{B}(E)$
- $\forall f \in \mathcal{B}(E)$ with $0 \leq f \leq 1$: $0 \leq P_t f \leq 1$
- $P_{s+t} = P_s \circ P_t$
- $\forall (f_n)_{n\in\mathbb{N}} \subseteq \mathcal{B}(E)$ with $f_n \searrow 0$: $P_t f_n \searrow 0$.

A sub-Markov semigroup is called normal if $P_0 = I$ and conservative if $\forall t \geq 0$: $P_t 1 = 1$.

Theorem 7.15

Let P be sub Markov transition function on $(E, \mathcal{B}(E))$ and define $\forall f \in \mathcal{B}(E), x \in E, t \geq 0$:

$$(P_t f)(x) := \int_E f(y) P(t, x, dy).$$

Then, $(P_t)_{t\geq 0}$ is a sub Markov semigroup. If additionally, P is normal then $(P_t)_{t\geq 0}$ is also normal. If P is Markov transition function, then $(P_t)_{t\geq 0}$ is conservative.

Definition 7.16

An adapted $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$-valued stochastic process X on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ is called Levy process if

- $\mathbb{P}(X_0 = 0) = 1$
- X has independent increments, i.e.

$$\forall 0 \leq s < t < \infty : \quad \sigma(X_t - X_s : t > s) \text{ and } \mathcal{F}_s \text{ are independent.}$$

- X has stationary increments:

$$\forall 0 \leq s \leq t < \infty : \quad \mathbb{P}_{X_t - X_s} = \mathbb{P}_{X_{t-s}}.$$

- X has Cadlag paths, i.e. $\forall \omega \in \Omega, t \geq 0$: $X_t(\omega) = X_{t+}(\omega) := \lim_{s \searrow t} X_s(\omega)$ and for all $t > 0$

$$X_{t-}(\omega) := \lim_{s \nearrow t} X_s(\omega).$$

  left limits exist and are finite.

Example 7.1

Let $A : B \to B$ be a bounded linear operator and define

$$P_t := e^{tA} := u - \lim_{n\to\infty} \sum_{k=1}^{n} \frac{t^k}{k!} A^k, \quad \forall t \geq 0.$$

Because $\forall n \in \mathbb{N}$ and $t \geq 0$:

$$\|\sum_{k=1}^{n} \frac{t^k}{k!} A^k\| \leq \sum_{k=1}^{n} \frac{t^k}{k!} \|A^k\| \leq \sum_{k=1}^{n} \frac{t^k}{k!} \|A\|^k \leq e^{t\|A\|} < \infty,$$

as A is bounded, yielding that $(P_t)_{t\geq 0}$ is well-defined. Additionally, $P_0 = A^0 = I$ and $\forall s, t \geq 0$:

$$\begin{aligned} P_{s+t} &= \sum_{k=0}^{\infty} \frac{(s+t)^k}{k!} A^k = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{l=0}^{k} \frac{k!}{l!(k-l)!} s^l t^{k-l} A^k \\ &= \sum_{l=0}^{\infty} \frac{1}{l!} s^l A^l \sum_{k=l}^{\infty} \frac{1}{(k-l)!} t^{k-l} A^{k-l} \\ &\overset{indexshift}{=} (\sum_{l=0}^{\infty} \frac{1}{l!} s^l A^l)(\sum_{k=0}^{\infty} \frac{1}{k!} t^k A^k) = P_s \cdot P_t. \end{aligned}$$

Thus, $(P_t)_{t\geq 0}$ is a semigroup. Additionally,

$$\|P_t - I\| \overset{???}{=} \|\sum_{k=1}^{\infty} \frac{t^k}{k!} A^k\| \leq \sum_{k=1}^{\infty} \frac{t^k}{k!} \|A\|^k = e^{t\|A\|} - 1 \to 0, \quad t \searrow 0.$$

Thus, $(P_t)_{t\geq 0}$ is uniformly continuous. Finally, $\forall t > 0$:

$$\|\frac{1}{t}(P_t - I) - A\| = \|\sum_{k=2}^{\infty} \frac{t^{k-1}}{k!} A^k\| \leq \sum_{k=2}^{\infty} \frac{t^{k-1}}{k!} \|A\|^k = \frac{1}{t}(e^{t\|A\|} - 1 - t\|A\|).$$

Because

$$\|\frac{1}{t}(P_t f - f) - Af\| \leq \|\frac{1}{t}(P_t - I) - A\|_{OP} \|f\| \to 0, t \searrow 0$$

it follows

$$s - \lim_{t \searrow 0} \frac{1}{t}(P_t f - f) = Af.$$

Example 7.2

We will derive the generator for the Brownian transition function P^{BM} on $\mathbb{R}^d$ for the respective semigroup $(P_t)_{t\geq 0}$ on $C_0(\mathbb{R}^d)$. Define

$$C_0^2(\mathbb{R}^d) \equiv C_0^2(\mathbb{R}^d, \mathbb{R} := \{f \in C_0(\mathbb{R}^d) \cap C^2(\mathbb{R}^d) \mid \partial_{ij}^2 f \in C_0(\mathbb{R}^d), i, j = 1, ..., d\}.$$

For $f \in C_0^2(\mathbb{R}^d)$ with Taylor expansion we get

$$f(x+y) = f(x) + \sum_{i=1}^{d} \partial_i f(x) y_i + \int_0^1 (1-\theta) \sum_{i,j=1}^{d} \partial_{ij}^2 f(x - \theta y) y_i y_j d\theta.$$

Then,

$$\begin{aligned}\frac{1}{t}((P_tf)(x)-f(x)) &= \frac{1}{t}(\mathbb{E}_x[f(B_t)]-f(x)) = \frac{1}{t}\mathbb{E}[f(x+B_t)-f(x)]\\ &= 0+\frac{1}{2t}\sum_{i,j=1}^{d}(\partial_{ij}^2 f)(x)\mathbb{E}[B_t^iB_t^j]\\ &+\sum_{i,j=1}^{d}\int_0^1(1-\theta)\mathbb{E}[(\partial_{ij}^2f(x-\theta B_t)-\partial_{ij}^2f(x))\frac{1}{t}B_t^iB_t^j]d\theta\\ &=: \frac{1}{2}(\Delta f)(x)+\sum_{i,j=1}^{d}\int_0^1(1-\theta)R_{ij}(x,\theta,t)d\theta\end{aligned}$$

To show: $\forall x\in\mathbb{R}^d$: $\sum_{i,j=1}^d\int_0^1(1-\theta)R_{ij}(x,\theta,t)d\theta \overset{u}{\to} 0$, $t\searrow 0$.
Because $\partial_{ij}^2 f$ is uniformly continuous, we get

$$\begin{aligned}|R_{ij}(x,\theta,t)| &\le \mathbb{E}[|\partial_{ij}^2f(x+\theta B_t)-\partial_{ij}^2f(x)|\frac{1}{t}|B_t^iB_t^j|]\\ &\overset{???}{\le} k^2\underbrace{\sup_{|z-z'|\le k\sqrt{t}}|\partial_{ij}^2f(z)-\partial_{ij}^2f(z')|}_{\to 0,t\searrow 0}+2\|\partial_{ij}^2f\|_\infty\underbrace{\mathbb{E}[\|B_1\|^2\mathbf{1}_{\|B_1\|>k}]}_{\to 0,k\to\infty}.\end{aligned}$$

Thus,

$$\sup_{x\in\mathbb{R}^d,\theta\in[0,1]}|R_{ij}(x,\theta,t)|\to 0, t\searrow 0, k\to\infty.$$

Then for every $f\in C_0^2(\mathbb{R}^d)$ we have that $Gf=\frac{1}{2}\Delta f$ and $C_0^2(\mathbb{R}^d)\overset{???}{\subseteq}\mathcal{D}(G)$.

Remark 7.3

- For $d=1$: $\mathcal{D}(\Delta)=C_0^2(\mathbb{R})$
- For $d>1$ the Laplace operator Δ on $C_0^2(\mathbb{R}^d)$ is not closed in $C_0(\mathbb{R}^d)$, yielding that $C_0^2(\mathbb{R}^d)\subset\mathcal{D}(\Delta)$.
- $(X_t)_{t\ge 0}$ is a Markov process $\iff$ $P(t,x,A)$ $\iff$ $(P_t)_{t\ge 0}$ $\implies$ $(G,\mathcal{D}(G))$.

Definition 7.17

Let A be a closed operator on the Banach space B with definition space $\mathcal{D}(A)$, then

$$\rho(A):=\{\lambda\in\mathbb{R}\mid(\lambda I-A)^{-1}\text{ exists and is bounded, linear operator}\}$$

is called resolvent set of A. $\sigma(A):=\mathbb{R}\setminus\rho(A)$ is called spectrum of A.

Remark 7.4

$\rho(A)$ is open subset and $\sigma(A)$ is closed.

Definition 7.18

Let A be closed linear operator on B with definition space $\mathcal{D}(A)$. For $\lambda \in \rho(A)$ the operator

$$R_\lambda := R_\lambda(A) := (\lambda I - A)^{-1}$$

is called resolvent of A.

Lemma 7.19

Let $(A, \mathcal{D}(A))$ be closed operator on B, then

- $\forall \lambda, \mu \in \rho(A) : R_\lambda - R_\mu = (\mu - \lambda) R_\lambda R_\mu$ and $R_\lambda R_\mu = R_\mu R_\lambda$.
- The image of R_λ is independent of $\lambda \in \rho(A)$, meaning that

  $$\forall \lambda, \mu \in \rho(A) : \quad R_\lambda B = R_\mu B.$$

  Additionally, $R_\lambda B = \mathcal{D}(A)$.
- If $(0, \infty) \subseteq \rho(A)$ and $\forall \lambda > 0$ with $\|\lambda R_\lambda\| \leq 1$ then

  $$\overline{R_\lambda B} = \{f \in B \mid \lambda R_\lambda f \xrightarrow{s} f, \lambda \to \infty\}.$$

Definition 7.20

Let $(P_t)_{t \geq 0}$ a contracting semigroup in B with respective generator $(G, \mathcal{D}(G))$ we call the resolvent $R_\lambda(G)$ for $\lambda \in \rho(G)$ of G in B_0 the resolvent of the semigroup $(P_t)_{t \geq 0}$.

Theorem 7.21

Let $(P_t)_{t \geq 0}$ be contracting semigroup on B and denote by $(G, \mathcal{D}(G))$ the respective generator. Then

- $\forall \lambda > 0 \; \exists R_\lambda(G)$ resolvent on B_0 and is equal to the Laplace transform L_λ of $(P_t)_{t \geq 0}$, i.e.

  $$\forall f \in B_0, \lambda > 0 : \quad R_\lambda f = L_\lambda f := \int_0^\infty e^{-\lambda s} P_s f ds.$$

  Especially, $\forall f \in B_0, \lambda > 0$: $\|\lambda R_\lambda f\| \leq \|f\|$.
- $\overline{R_\lambda B_0} = B_0$.

Remark 7.5

For $R_\lambda : B_0 \to \mathcal{D}(G)$ its inverse is given by $\lambda Id - G$.

Example 7.3

We want to find the generator for the 1-dim. Brownian motion on $C_0(\mathbb{R})$ via the resolvent. With the previous Lemma it follows for every $\lambda > 0$:

$$R_\lambda C_0(\mathbb{R}) = \mathcal{D}(G) = \{R_\lambda g \mid g \in C_0(\mathbb{R})\} \quad \text{and} \quad Gf = (\lambda Id - R_\lambda^{-1}) f = \lambda f - R_\lambda^{-1} f.$$

For $g \in C_0(\mathbb{R})$:

$$\begin{aligned} f(x) = R_\lambda g(x) = L_\lambda g(x) &= \int_0^\infty e^{-\lambda s} P_s g ds \\ &= \int_0^\infty e^{-\lambda s} \int_{\mathbb{R}} g(y) \frac{1}{\sqrt{2\pi s}} e^{-\frac{(x-y)^2}{2s}} dy ds \\ &= \int_{\mathbb{R}} g(y) \int_0^\infty \frac{1}{\sqrt{2\pi s}} e^{-\lambda s - \frac{x-y}{2s}} ds dy \\ &= \int_{\mathbb{R}} g(y) \frac{1}{\sqrt{2\lambda}} e^{-\sqrt{2\lambda}(x-y)} dy \\ &= \frac{e^{-\sqrt{2\lambda}x}}{\sqrt{2\lambda}} \int_{-\infty}^x g(y) e^{\sqrt{2\lambda}y} dy + \frac{e^{\sqrt{2\lambda}x}}{\sqrt{2\lambda}} \int_x^\infty g(y) e^{-\sqrt{2\lambda}y} dy. \end{aligned}$$

Then $f \in C_0(\mathbb{R})$ and is twice differentiable:

$$\begin{aligned} f'(x) &= e^{-\sqrt{2\lambda}x} \int_{-\infty}^x g(y) e^{\sqrt{2\lambda}y} dy + e^{\sqrt{2\lambda}x} \int_x^\infty g(y) e^{-\sqrt{2\lambda}y} dy \\ f''(x) &= \sqrt{2\lambda} e^{-\sqrt{2\lambda}x} \int_{-\infty}^x g(y) e^{\sqrt{2\lambda}y} dy + \sqrt{2\lambda} e^{\sqrt{2\lambda}x} \int_x^\infty g(y) e^{-\sqrt{2\lambda}y} dy - 2g(x) \\ &= 2(\lambda R_\lambda g(x) - g(x)) = 2(\lambda f(x) - g(x)) \\ &\iff \frac{1}{2} f'' = \lambda f - g = \lambda f - R_\lambda^{-1} f = Gf \\ &\iff Gf = \frac{1}{2} f''. \end{aligned}$$

So $f'' \in C_0(\mathbb{R})$ and thus

$$\mathcal{D}(G) \subseteq \{f \in C_0(\mathbb{R} \cap C^2(\mathbb{R})) \mid f'' \in C_0(\mathbb{R})\} =: A.$$

Now we show equality. Let $f \in A$ define

$$g := R_\lambda(\lambda f - \frac{1}{2} f'') \in R_\lambda C_0(\mathbb{R}).$$

Set $h := f - g$, then $\forall \lambda > 0$:

$$\lambda h - \frac{1}{2} h'' = \lambda f - \frac{1}{2} f'' - (\lambda Id - G) R_\lambda (\lambda f - \frac{1}{2} f'') = 0.$$

Thus, $h = 0$, as the above holds for every $\lambda > 0$. We get $f = g = R_\lambda(\lambda f - \frac{1}{2} f'')$ which implies that $f \in R_\lambda C_0(\mathbb{R}) = \mathcal{D}(G)$. Thus $\mathcal{D}(G) = A$.

Theorem 7.22

A strong continuous semigroup linear contractions $(P_t)_{t \geq 0}$ is uniquely determined by its respective generator $(G, \mathcal{D}(G))$.

Theorem 7.23

Let G be linear operator on B with definition space $\mathcal{D}(G)$. G is generator of a strong continuous semigroup $(P_t)_{t\geq 0}$ on $B \iff$

- $\overline{\mathcal{D}(G)} = B(\equiv B_0)$
- $(0,\infty) \in \rho(G)$ and $\forall \lambda > 0$: $\|\lambda R_\lambda\| \leq 1$.

Is there a simpler to check criterion instead of ii) from the above for the reconstruction of a strong continuous semigroup linear contractions from the generator?

Theorem 7.24

Let E be a compact space and G a linear operator on $C(E)(C_b(E) = C_0(E))$ with definition space $\mathcal{D}(G)$. G is generator of a strong, continuous and conservative sub-Markovian semigroup $(P_t)_{t\geq 0}$ linear contractions on $C(E) \iff$

- $\overline{\mathcal{D}(G)} = C(E)(= B_0)$
- $\forall \lambda > 0, g \in C(E) : \lambda f - Gf = g$ has at least one solution in $\mathcal{D}(G)$
- Maximum principle: $\forall f \in \mathcal{D}(G)$ with $f(x_0) := \max_{x\in E} f(x)$ holds

$$(Gf)(x_0) \leq 0$$

- $1 \in \mathcal{D}(G)$ and $G1 = 0$.

Definition 7.25

A process is called progressively measurable if $\forall t \geq 0$:

$$[0,t] \times \Omega \to E, \quad (s,\omega) \mapsto X_s(\omega)$$

is $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$ measurable.

This implies that X is $\mathcal{F}_t$ adapted.

Theorem 7.26

Let X be progressively measurable, $(E,\mathcal{E})$ valued Markov process on $(\Omega,\mathcal{F},(\mathcal{F}_t)_{t\geq 0})$ wrt. the distribution $(\mathbb{P}_x)_{x\in E}$ with generator $(G,\mathcal{D}(G))$, i.e. G is generator on $B_0 \subseteq B(E)$ restricted to the respective semigroup $(P_t)_{t\geq 0}$, then

- $\forall x \in E, f \in \mathcal{D}(G), t \geq 0$:

$$M_t^f := f(X_t) - f(X_0) - \int_0^t (Gf)(X_r)dr$$

  is a $(\mathcal{F}_t)$-martingale

- $\forall x \in E, \lambda > 0, f \in R_\lambda B_0, t \geq 0$:

$$M_t^{f,\lambda} := e^{-\lambda t}(R_\lambda f)(X_t) - (R_\lambda f)(X_0) + \int_0^t e^{-\lambda r} f(X_r)dr$$

is a $(\mathcal{F}_t)$-martingale.

Definition 7.27

A $\mathcal{F}$-measurable mapping $T : \Omega \to [0, \infty]$ is called

- $(\mathcal{F}_t)$ optional time if $\forall t \geq 0$: $\{T \leq t\} \in \mathcal{F}_t^+ := \bigcap_{\epsilon>0} \mathcal{F}_{t+\epsilon}$.
- $(\mathcal{F}_t)$ stopping time if $\forall t \geq 0$: $\{T \leq t\} \in \mathcal{F}_t$.

Theorem 7.28

Let X be a strong Markov process on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, (\mathbb{P}_x)_{x\in E})$ with values in the polish space $(E, B(E))$. Denote by $(G, \mathcal{D}(G))$ the generator of the respective semigroup $(P_t)_{t\geq 0}$ on $B_0 \subseteq B(E)$. Then for every $f \in \mathcal{D}(G)$ and every $\mathcal{F}_t$ optional time S with $\mathbb{E}_x[S] < \infty$, where $x \in E$ we have

$$\mathbb{E}_x[f(X_S)] = f(x) + \mathbb{E}_x[\int_0^S (Gf)(X_r)dr].$$

Dirichlet Problem:
Let $D \subseteq \mathbb{R}^d$ be open and bounded with smooth boundary, $g : \overline{D} \to \mathbb{R}$, $\varphi : \partial D \to \mathbb{R}$ continuous. Find $f \in C^2(C) \cap C(\overline{D})$:

$$\frac{1}{2}\Delta f = g, \quad \text{on } D \quad \text{and} \quad f = \varphi \text{ on } \partial D.$$

Theorem 7.29

For $f \in C^2(D) \cap C(\overline{D})$ solution to the Dirichlet problem, $(B_t)_{t\geq 0}$ a d-dimensional Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, (\mathbb{P}_x)_{x\in\mathbb{R}^d})$ and $T_{D^C} := \inf\{t \geq 0 \mid B_t \notin D\}$ it holds $\forall x \in \bar{D}$:

$$f(x) = \mathbb{E}_x[\varphi(B_{T_{D^C}}) + \int_0^{T_{D^C}} g(B_r)dr]$$

and f is uniquely determined.

Proof. As D^C is closed it follows that T_{D^C} is $(\mathcal{F}_t)$ stopping time.
To show: $\forall x \in \mathbb{R}^d$: $\mathbb{E}_x[T_{D^C}] < \infty$.
Define the Lyaponov function

$$h(x) := -\underbrace{2e^{-\min_{x\in\bar{D}} x_1}}_{=:\mu} e^{x_1},$$

then

$$(Gh)(x) = \frac{1}{2}(\Delta h)(x) = -\frac{\mu}{2}e^{x_1} \leq -\frac{\mu}{2}e^{\min_{x\in\mathcal{D}} x_1} = -1.$$

For the bounded optimal time $T_{D^C} \wedge t$ and $x \in D$ we have

$$\mathbb{E}_x[T_{D^C} \wedge t] \overset{???}{\leq} \mathbb{E}_x[\int_0^{T_{D^C}\wedge t} (Gh)(B_r)dr]$$
$$\overset{Thm.Above}{=} h(x) - \mathbb{E}_x[h(B_{T_{D^C}\wedge t})] \overset{???}{\leq} 2\max_{y\in\bar{D}} |h(y)|.$$

Thus, with Lemma of Fatou we have $\forall x \in D$:

$$\mathbb{E}_x[T_{D^C}] \leq \liminf_{t\to\infty} \mathbb{E}_x[T_{D^C}\wedge] \leq 2\max_{y\in\bar{D}} |h(y)| < \infty.$$

Because $\mathbb{P}_x(T_{D^C} = 0) = 1$ for every $x \in D^C$, it follows that for every $x \in \mathbb{R}^d$ that $\mathbb{E}_x[T_{D^C}] < \infty$. Thus, for every $x \in \bar{D}$:

$$\begin{aligned} f(x) &= \mathbb{E}_x[\underbrace{f(B_{T_{D^C}})}_{=\varphi(B_{T_{D^C}})} - \int_0^{T_{D^C}} \underbrace{(Gh)(B_r)}_{=-g(B_r)} dr] \\ &= \mathbb{E}_x[\varphi(B_{T_{D^C}}) + \int_0^{T_{D^C}} g(B_r) dr]. \end{aligned}$$

□

Remark 7.6

$u : [0, \infty) \to D(G)$ is a solution of the iVP if

$$\frac{d}{dt}u(t) := s - \lim_{h \searrow} \frac{u(t+h) - u(t)}{h},$$

exists and strong continuous in t.

Theorem 7.30

Let $(P_t)_{t \geq 0}$ be semigroup linear contractions on a Banach space B with generator $(G, D(G))$. Additionally, let $f \in D(G)$ and $g : [0, \infty) \to B_0$ continuously differentiable in strong sense. Then the iVP

$$\frac{d}{dt}u(t) = Gu(t) + g(t), \quad t \geq 0, \quad u(0) = f$$

has exactly one solution

$$u : [0, \infty) \to D(G), \quad t \mapsto u(t) = P_t f + \int_0^t P_{t-s} g_s ds.$$

Proof. • Uniqueness: Assume u_1 and u_2 are two solutions and define $v := u_1 - u_2 : [0, \infty) \to D(G)$ Then it solves the iVP

$$\frac{d}{dt}v(t) = Gv(t), \quad t \geq 0, \quad v(0) = 0.$$

To show: $\|v(t)\| = 0$, $t \geq 0$.
Fix $T > 0$. Because $Gu : [0, \infty) \to B$ is strongly continuous the mapping $[0, T] \ni t \mapsto Gv(t)$ is uniformly strongly continuous, i.e.

$$\forall \epsilon > 0 \exists \delta > 0 \forall s, t \in [0, T] \text{ with } |s - t| < \delta : \quad \|Gv(t) - Gv(s)\| \leq \frac{\epsilon}{T}.$$

Choose $0 = t_0 < ... < t_n = T$ a partition of $[0, T]$ with $|t_k - t_{k-1}| < \delta$, $k = 1, ..., n$, then due to

$u(t) \in D(G) = R_\lambda B_0$, $\lambda > 0$, it follows that

$$\begin{aligned}
\|v(T)\| - \|v(0)\| &= \sum_{k=1}^{n} \|v(t_k)\| - \|v(t_{k-1})\| \\
&= \sum_{k=1}^{n} \| \underbrace{R_{(t_k - t_{k-1})^{-1}} \frac{1}{t_k - t_{k-1}} (Id - G)}\, v(t_k)\| - \|v(t_{k-1})\| \\
&\leq \sum_{k=1}^{n} \underbrace{\|R_{(t_k - t_{k-1})^{-1}}\|}_{\leq (t_k - t_{k-1})^{-1}} \| \frac{1}{t_k - t_{k-1}} (Id - G) v(t_k)\| - \|v(t_k) - \int_{t_{k-1}}^{t_k} Gv(t_k) ds\| \\
&\leq \sum_{k=1}^{n} \|v(t_k) - \int_{t_{k-1}}^{t_k} Gv(t_k) ds\| - \|v(t_k) - \int_{t_{k-1}}^{t_k} Gv(s) ds\| \\
&\overset{\|A\| - \|B\| \leq \|A - B\|}{\leq} \sum_{k=1}^{n} \int_{t_{k-1}}^{t_k} \|Gv(s) - Gv(t_k)\| ds < \epsilon.
\end{aligned}$$

Because by assumption $\|v(0)\| = 0$ and $\epsilon > 0$ the assertion follows.

- Existence:
  Consider $\forall t \geq 0$: $u(t) := P_t f + \int_0^t P_{t-s} g(s) ds$.
  To show: u solves the iVP.
  Because $f \in D(G)$ and $P_t f \in D(G)$ and $G P_t f = P_t G f$ it follows

$$\begin{aligned}
\int_0^t P_{t-s} g(s) ds &= \int_0^t P_{t-s} (g(0) + \int_0^s g'(r) dr) ds \\
&= \int_0^t P_{t-s} g(0) ds + \int_0^t \int_r^t P_{t-s} g'(r) ds dr \\
&= \underbrace{\int_0^t P_s g(0) ds}_{\in D(G)} + \int_0^t \underbrace{\int_0^{t-r} P_s g'(r) ds}_{\in D(G), \forall r \in [0,t]}\, dr.
\end{aligned}$$

  Thus, for all $t \geq 0$: $u(t) \in D(G)$. Additionally,

$$\begin{aligned}
\frac{d}{dt} u(t) &= \frac{d}{dt} P_t f + \frac{d}{dt} \int_0^t P_{t-s} g(s) ds \\
&= G P_t f + \frac{d}{dt} (\int_0^t P_s g(0) ds + \int_0^t \int_0^{t-r} P_s g'(r) ds dr) \\
&\overset{???}{=} \underbrace{G P_t f}_{= Gu(t) - G(\int_0^t P_s g(0) ds + \int_0^t \int_0^{t-r} P_s g'(r) ds dr)} + P_t g(0) + \int_0^t P_{t-r} g'(r) dr
\end{aligned}$$

  The theorem from stated that $P_t f - f = G(\int_0^t P_s f ds)$ yields that

$$\begin{aligned}
\frac{1}{h}(P_h - Id) \int_0^t \int_0^{t-r} P_s g'(r) ds dr &= \int_0^t \frac{1}{h} (\int_h^{t-r+h} P_s g'(r) ds - \int_0^{t-r} P_s g'(r) ds) dr \\
&= \int_0^t \underbrace{\frac{1}{h} \int_{t-r}^{t-r+h} P_s g'(r) ds}_{\to P_{t-r} g'(r), h \searrow 0} - \underbrace{\frac{1}{h} \int_0^h P_s g'(r) ds}_{\to g'(r), h \searrow 0}\, dr.
\end{aligned}$$

Thus, with the theorem of Lebesgue it follows

$$\begin{aligned} G\left(\int_0^t \int_0^{t-r} P_s g'(r) ds dr\right) &= \lim_{h \searrow 0} \frac{1}{h}(P_h - Id) \int_0^t \int_0^{t-r} P_s g'(r) ds dr \\ &\overset{above}{=} \int_0^t P_{t-r} g'(r) dr - \int_0^t g'(r) dr \\ &= \int_0^t P_{t-r} g'(r) dr - (g(t) - g(0)). \end{aligned}$$

Hence, finally

$$\begin{aligned} \frac{d}{dt} u(t) &= Gu(t) - G\left(\int_0^t P_s g(0) ds + \int_0^t \int_0^{t-r} P_s g'(r) ds dr\right) + P_t g(0) + \int_0^t P_{t-r} g'(r) dr \\ &= Gu(t) + g(t) + g(0) - P_t g(0) \overset{???}{=} Gu(t) + g(t). \end{aligned}$$

□

Corollary 7.31

Let X be progressively measurable Markov process on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, (\mathbb{P}_x)_{x\in E})$ with values in the polish space $(E, \mathcal{B}(E))$. Additionally, $(G, D(G))$ generates the respective semigroup $(P_t)_{t\geq 0}$ on $B_0 \subseteq B(E)$. Then for $f \in D(G)$ and $g : [0, \infty) \to B_0$ continuously differentiable in the strong sense, then

$$u(t, x) = \mathbb{E}_x[f(X_t) + \int_0^t g(t - s, X_s) ds]$$

is the unique solution of the iVP

$$\frac{d}{dt} u(t) = Gu(t) + g(t), t \geq 0, \quad u(0) = f.$$

Proof. Because $g : [0, \infty) \times E \to \mathbb{R}$ is continuous in the first argument and measurable in the second, it follows that g is $(\mathcal{B}([0, \infty)) \otimes \mathcal{B}(E), \mathcal{B}(\mathbb{R}))$ measurable and by assumption bounded. Thus, the integral is well-defined. Finally,

$$\begin{aligned} u(t, x) &= (P_t f)(x) + \int_0^t P_{t-s} g(s, x) ds = \mathbb{E}_x[f(X_t)] + \int_0^t \mathbb{E}_x[g(s, X_{t-s})] ds \\ &= \mathbb{E}_x[f(X_t)] + \int_0^t \mathbb{E}_x[g(t - s, X_s)] ds \\ &= \mathbb{E}_x[f(X_t) + \int_0^t g(t - s, X_s) ds]. \end{aligned}$$

□

SECTION 8

BACKWARD MARKOV PROCESSES

SECTION 9

TIME-REVERSAL OF DIFFUSIONS

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